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Fast and Stable Virtual Human Video Generation: A Survey

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Abstract Deploying virtual-human video generation requires timely delivery and consistent identity, appearance, and motion over sustained sessions. This survey examines the field through two perspectives, *Fast* and *Stable*, focusing on talking portraits and selected upper- and full-body systems. We classify methods into generative adversarial network (GAN), diffusion, visual-autoregressive, and rendering routes by the component that produces appearance, while separating control, state, and decoding in hybrid pipelines. We review acceleration through compact representations, fewer model evaluations, computation reuse, and system optimization, and examine how reference anchors, bounded memory, position re-indexing, history-robust training, and boundary-state transfer mitigate temporal drift. We complement the review with runtime measurements of selected methods and performance evaluations on HDTF and StreamEval. Finally, we review datasets, evaluation protocols, and open challenges.

1 Introduction

Virtual-human video generation synthesizes a subject whose appearance and behavior follow speech, text, pose, expression, or video inputs. A reference image, enrollment recording, or learned subject representation specifies identity, while the output covers the face, upper body, or full body. Across the tasks in Figure 1, the common objective is to preserve subject appearance while reproducing the requested motion [145, 200, 239, 266].

Interactive applications and client-side or edge deployment impose two coupled requirements on virtual-human generation. Virtual assistants, customer support, telepresence, and live communication need timely output within a latency and resource budget [68, 87, 233]. They also need consistent identity, appearance, synchronization, and motion throughout a session [123, 179]. We call these perspectives *Fast* and *Stable*. Figure 2 illustrates playback deadlines and identity preservation under changing expression and pose. High frame throughput alone establishes neither low startup latency nor resistance to gradual degradation. These properties require separate runtime accounting and time-resolved stability measurements over a declared duration.

Recent methods pursue efficiency and consistency through different pipeline stages (Figure 3). Step distillation shortens diffusion trajectories [138, 189, 251, 252]. Feature and key-value (KV) caching reuses computation across denoising steps or adjacent frames [128, 140, 143]. Visual-autoregressive (AR) models predict frame state from cached causal history [28, 111, 231], while rendering reuses enrolled appearance across driving sequences [68, 69, 227, 233]. Accelerating synthesis can shift the bottleneck to conditioning, decoding, or composition. We therefore organize methods by the module directly responsible for producing appearance, termed the *visual owner*, and examine its costs and temporal state within the complete pipeline.

1.1 Scope: Fast and Stable

This survey covers human-centered RGB video generators that reduce runtime cost or support extended generation. We focus on talking portraits and selected upper- and full-body systems in both one-shot and subject-specific settings. Neural radiance fields (NeRFs), 3D Gaussian splatting (3DGS), and other rendering backends are included when they enable efficient appearance construction or frame synthesis [99, 149]. Predictors that output only meshes, 3D morphable model (3DMM) coefficients, or skeletal motion are treated as supporting controllers within RGB pipelines.

Fast covers the work and wall-clock time needed to deliver visible frames: representation size, strict number of function evaluations (NFE), per-call complexity, computation reuse, and decoding or rendering overhead. **Stable** covers visual and behavioral consistency over a continuous evaluated horizon. It examines drift and the anchoring, memory, training, and recovery policies used to limit error accumulation. We record timing boundaries, hardware, demonstrated durations, and quantitative drift evidence separately because improving one property can compromise the other.

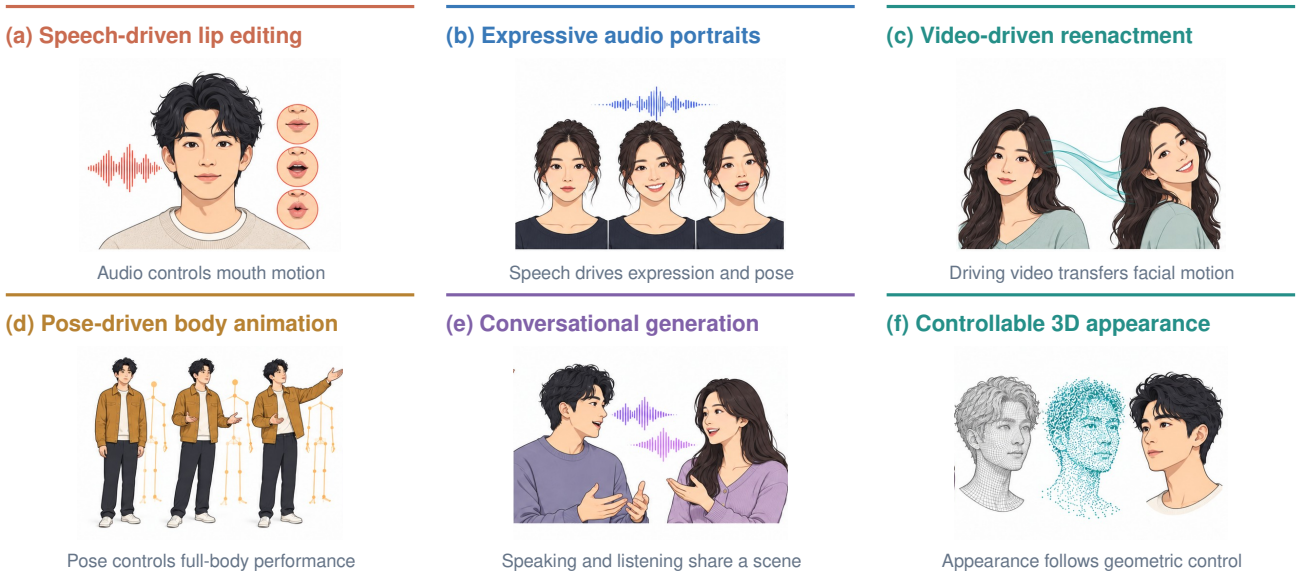


Fig. 1 Schematic examples of virtual-human generation tasks: (a) speech-driven lip editing, (b) expressive audio portraits, (c) video-driven reenactment, (d) pose-driven body animation, (e) conversational generation, and (f) controllable 3D appearance. The illustrations indicate task scope and controls.

Interactivity is the ability to adapt online to changing inputs, including dialogue turns, interruptions, or abrupt pose shifts [87, 123, 242, 258, 263]. An interactive generator can still accumulate identity drift or need state resets. We review its acceleration and state-management mechanisms, while dialogue-quality assessment belongs to dedicated interaction surveys and benchmarks [8, 16, 53, 255]. Section 2 formalizes the system boundaries and evaluation terms.

1.2 Literature Selection and Reporting Conventions

We select studies that explain how appearance representations, computation reuse, or persistent state affect efficiency and sustained generation. Each primary-route entry must document a complete RGB pipeline with an identifiable visual owner. Foundational papers supply representation concepts, while contextual studies cover motion or dialogue interfaces. We distinguish releases and serving configurations when their architecture or runtime differs. To examine cost and quality under declared conditions, we measure open-source implementations in a single-H200 runtime benchmark (Section 8.3) and a matched-input HDTF quality benchmark (Section 8.4).

The overview includes representative non-real-time generators as well as systems with reported real-time operation. Background generators establish a representation or conditioning mechanism; their inclusion does not establish low latency or sustained stability. Figure 3 presents the complete 136-method timeline, while Figure 4 selects representative examples for the highlighted mechanism groups. References and classification decisions are discussed in the corresponding route chapters. Figure density is not a measure of the field's publication volume, and the illustrated methods are a different selection from the 51 measured runtime configurations. Neither figure is a matched-hardware performance ranking.

The 2021–2026 range follows cited editions rather than uniformly using first-preprint years. Within-year placement does not establish publication priority. A renamed work is counted once: SoulX-FlashTalk denotes the work initially released as SoulX-LiveTalk in the cited 2025 version, while AvatarSync retains its 2025 edition despite the later FluentAvatar name [52, 179]. The 2024 entries for Hallo3 and LatentSync likewise refer to their cited first preprints; the catalogue records the display names and citation keys together [44, 115].

1.3 Relation to Existing Surveys

Related surveys organize the field by different questions. Efficient-diffusion reviews cover distillation, caching, attention optimization, and serving [178], while long-video reviews examine context extension and memory costs [112]. Human-generation surveys use task stages, driving modalities, or 2D/3D representations [145, 170, 175, 239]. Meng et al. also discuss system integration, including identity drift, interfaces, and real-time constraints [145]. Graphics surveys examine avatar construction and appearance priors [293], and interactive-video

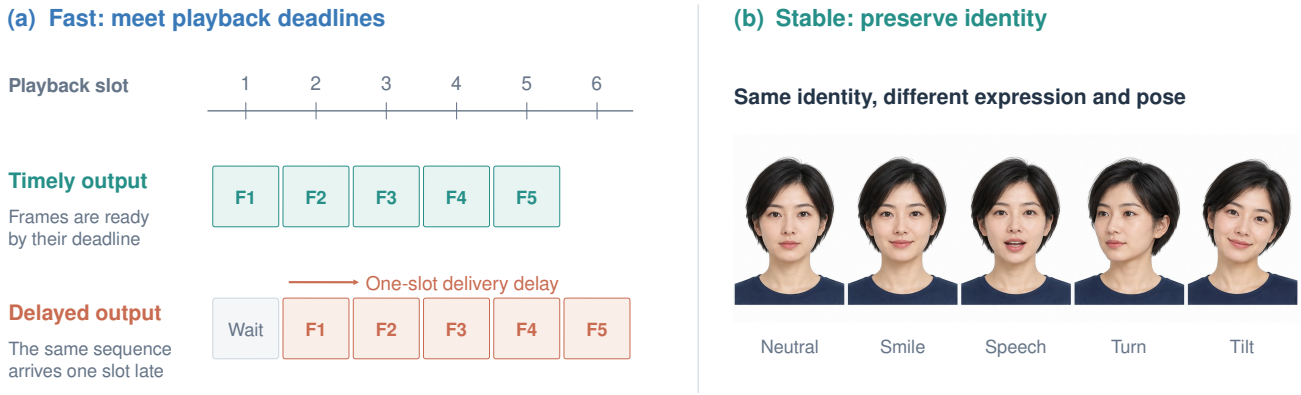


Fig. 2 Schematic Fast and Stable requirements. (a) The same output sequence meets playback deadlines or arrives one slot late. (b) Identity remains consistent while expression and pose follow the input. Both panels illustrate desired behavior.

and real-time-avatar reviews emphasize responsiveness and production constraints [53, 255].

Table 1 positions our analysis within these perspectives. We connect the visual owner to stage-wise cost and temporal state, recording auxiliary motion controllers and decoders separately in hybrid pipelines.

1.4 Ownership Taxonomy

We group methods into four visual-owner routes. Hybrid systems follow an ordered test of rendering, visual autoregression, diffusion, and GAN synthesis (Section 2.3). For example, a diffusion motion predictor driving a Gaussian avatar belongs to the render-based route.

- **GAN-based methods** (Section 3) employ an adversarially trained feed-forward generator to synthesize dense portraits or localized facial regions [64, 97, 164, 201, 268]. Cropped synthesis and compact motion bases reduce per-frame generator cost, leaving temporal coherence dependent on input conditioning and output blending.
- **Diffusion-based methods** (Section 4) iteratively denoise or integrate dense pixels, spatiotemporal latents, or video chunks [75, 79, 127, 200]. Runtime acceleration relies on latent compression, step distillation, caching, and multi-GPU pipelining, while streaming stability relies on reference sinks, bounded KV memory, and self-history training.
- **Autoregressive methods** (Section 5) natively predict evolving visual tokens or frame latents conditioned on prior visual state [5, 28, 225, 231]. Spatial token compression and parallel local emission heads mitigate serial latency, while KV window policies govern the trade-off between context retention and error propagation.
- **Render-based methods** (Section 6) query or deform a subject-specific mesh, neural radiance field, Gaussian scene, or identity-conditioned feature volume [68, 99, 119, 149, 233]. Reusing enrolled appearance amortizes synthesis cost and provides a structural identity anchor, shifting runtime and stability challenges to enrollment, tracking, deformation, and compositing.

Figure 3 contains 21 GAN, 58 diffusion, five visual-AR, and 52 rendering methods. Figure 4 uses a representative subset to explain 14 selected mechanism groups and their Fast/Stable implications. Conditioned diffusion generators are summarized as historical background in Section 4.1. The selection includes non-real-time baselines; the 2021 cutoff applies to the overview figures, not to foundational references or benchmark inclusion. Within each route, we compare driving modality, anatomical scope, personalization, representation, and serving topology independently. Auxiliary control and decoding modules retain their roles without duplicating a system across routes.

1.5 Contributions and Main Findings

We contribute a visual-ownership taxonomy for hybrid systems, a cross-route analysis of efficiency and stability, an evaluation review, and two benchmarks of open-source implementations. Section 7 develops four design findings:

1. **Decoder selection reshapes the speed–fidelity frontier.** Substituting a lightweight variational autoencoder (VAE) accelerates frame delivery after denoiser distillation, but can trade away identity similarity and temporal smoothness, demonstrating that post-denoiser decoding directly impacts both Fast and Stable metrics.
2. **Steady-state throughput gains do not automatically reduce startup latency.** Multi-GPU timestep pipelining raises steady-state frame

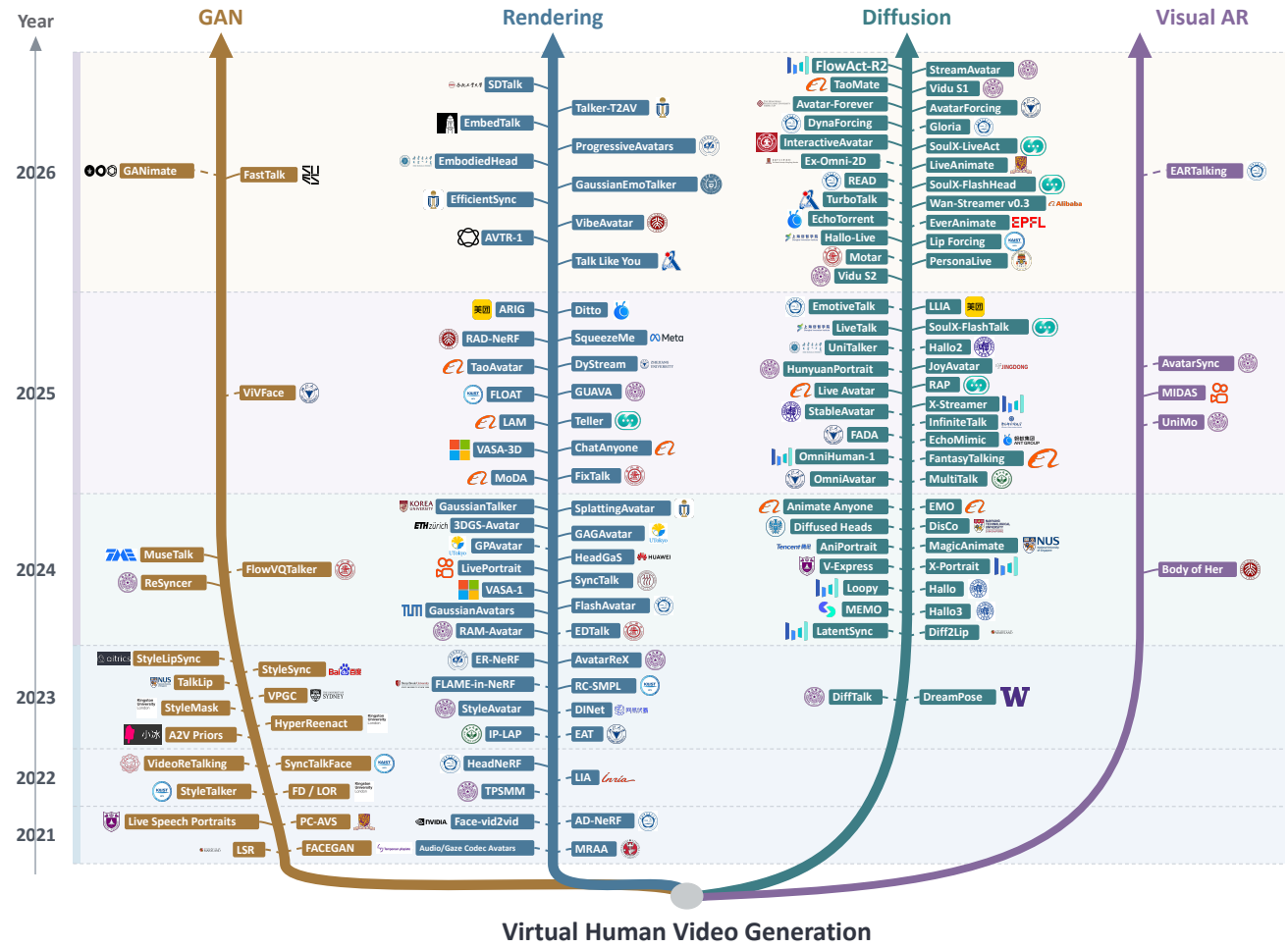


Fig. 3 Timeline of 136 human-video methods across four visual-owner routes. **Solid branches: claimed real-time performance; dashed branches: real-time capability not reported.** Years follow cited editions; positions within a year do not establish priority. Branches indicate grouping, not inheritance. Hardware and timing boundaries vary across reports. Mechanism groups and representative examples are shown in Figure 4.

throughput while adding pipeline-fill delay to the first emitted chunk, requiring dedicated decoder parallelism and kernel optimization to lower both metrics jointly.

3. **Bounded memory allocation is distinct from demonstrated temporal stability.** Enforcing a fixed-size KV cache or memory bank guarantees bounded hardware consumption, whereas preventing visual degradation over multi-minute rollouts requires empirical validation under non-stationary inputs.
4. **Architectural appearance reuse leaves non-identity failure paths active.** Maintaining a persistent 3D asset or source feature volume anchors identity by construction, yet pose drift, tracking jitter, and deformation wobble still arise along the control and rendering path.

We also conduct two empirical studies. The runtime study covers 51 single-H200 configurations: 8 GAN, 16 diffusion, 26 rendering, and one Faces Learned with an Articulated Model and Expressions (FLAME) motion-visualization baseline. Each runs three measured calls after three same-interpreter warm-ups. We separate first-call loading from warmed stage times and peak memory, distinguishing 10 resident configurations from 41 that reload weights (Section 8.3). The quality study evaluates 23 audio-driven configurations on 40 fifteen-second HDTF cases, producing 920 videos (Section 8.4). SyncNet, ArcFace, and unpaired full-frame FID/FVD expose trade-offs among lip synchronization, identity retention, and visual quality.

1.6 Organization

Section 2 defines system boundaries, ownership, cost, and stability. Sections 3–6 review the four routes, and Section 7 compares their trade-offs. Section 8 presents datasets, metrics, and our measurements. Section 9 ad-

Table 1 Complementary survey perspectives, emphasizing the tasks, organizing principles, and practical questions addressed by each review. Throughout the tables, Year denotes the year of the cited publication, including preprints.

Survey / scope	Year	Organizing principle	Efficiency and temporal emphasis	Relation to the present analysis
Deep Person Generation [175]	2023	Face, pose, and clothing generation, shared technical components	Identity/attribute control, generation quality, and task-specific challenges	Connects human-generation tasks, our comparison unit is the deployed RGB pipeline.
Talking Head Synthesis [145]	2026	Portrait generation, driving mechanisms, editing, and system integration	Synthesis and control; identity drift, interfaces, and real-time constraints	Shares system concerns; we organize hybrid RGB pipelines by visual owner and trace stage costs and persistent state.
Human Motion Video Generation [239]	2025	Input, motion planning, video generation, refinement, and output, modality and task	Motion/video methods, datasets, evaluation, and temporal challenges	Provides broad human-motion coverage, we focus on cost, state, enrollment, and runtime and temporal behavior.
Efficient Diffusion Models [178]	2025	Algorithm, system, and framework levels	Sampling and implementation efficiency across diffusion applications	Supplies acceleration concepts, we track which stage is accelerated and include GAN, visual AR, and rendering pipelines.
Interactive Real-Time Visual Avatar Generation [53]	2026	Method families and latency, anatomy, personalization, and input modality	Production constraints and head–body integration, drift as a challenge	Closely related deployment perspective, our analysis connects module roles with computation and temporal behavior.
This survey	2026	Visual owner plus independent control/state/decoder roles	Fast/Stable lenses, timing boundaries, enrollment, seven stability dimensions	Cross-route mechanism analysis, 51 single-H200 runtime/memory profiles and a 23-model, 920-video HDTF quality benchmark.

dresses interfaces, deployment choices, provenance, and open questions.

2 Preliminaries and Common Evaluation Language

This section defines the system boundaries, ownership rules, costs, and stability mechanisms used to compare heterogeneous pipelines.

2.1 Problem Formulation and System Boundary

Let $V = (I_1, \ldots, I_T)$ be a generated video of T frames. The pipeline takes enrollment data e, runtime conditions $c_{1:T}$, and persistent state h_t. Enrollment may use a portrait, monocular video, multiview capture, or avatar asset. Conditions may include speech, text, pose, expression, or driving video. At step t, the system emits a frame or a chunk of length k and updates its state:

$$(I_{t:t+k-1}, h_{t+k}) = \mathcal{G}(e, c_{\mathcal{V}_t}, h_t; \theta), \tag{1}$$

where $\mathcal{V}_t$ is the conditioning window consumed at that step. This formulation covers one-shot, subject-specific, and interactive systems without requiring reference re-encoding at each step. *Identity fidelity* preserves the enrolled appearance. *Driving fidelity* reproduces requested articulation and motion. *Temporal fidelity* maintains coherent transitions across frames and chunks.

To separate representation synthesis from pixel reconstruction in hybrid pipelines, we factor Equation (1) into an intermediate representation generator G and a decoding or rendering operator D:

$$\begin{aligned}(r_{t:t+k-1}, h_{t+k}) &= G(e, c_{\mathcal{V}_t}, h_t),\\ I_{t:t+k-1} &= D(r_{t:t+k-1}).\end{aligned} \tag{2}$$

Here, r is the intermediate control or visual state. Subject appearance is carried by r or encoded in the parameters of D. Although G and D may be fused, this decomposition separates enrollment from recurring inference and decoding costs (Figure 5).

2.2 Efficiency and Stability

Efficiency measures work and wall-clock cost across conditioning, synthesis, decoding or rendering, and composition. *Stability* measures identity, appearance, synchronization, and motion over a declared continuous horizon. Each requires appropriate evidence. A bounded KV cache establishes a memory ceiling, while visual drift requires time-resolved measurements. Enrollment duration and long qualitative demonstrations alone do not establish stability.

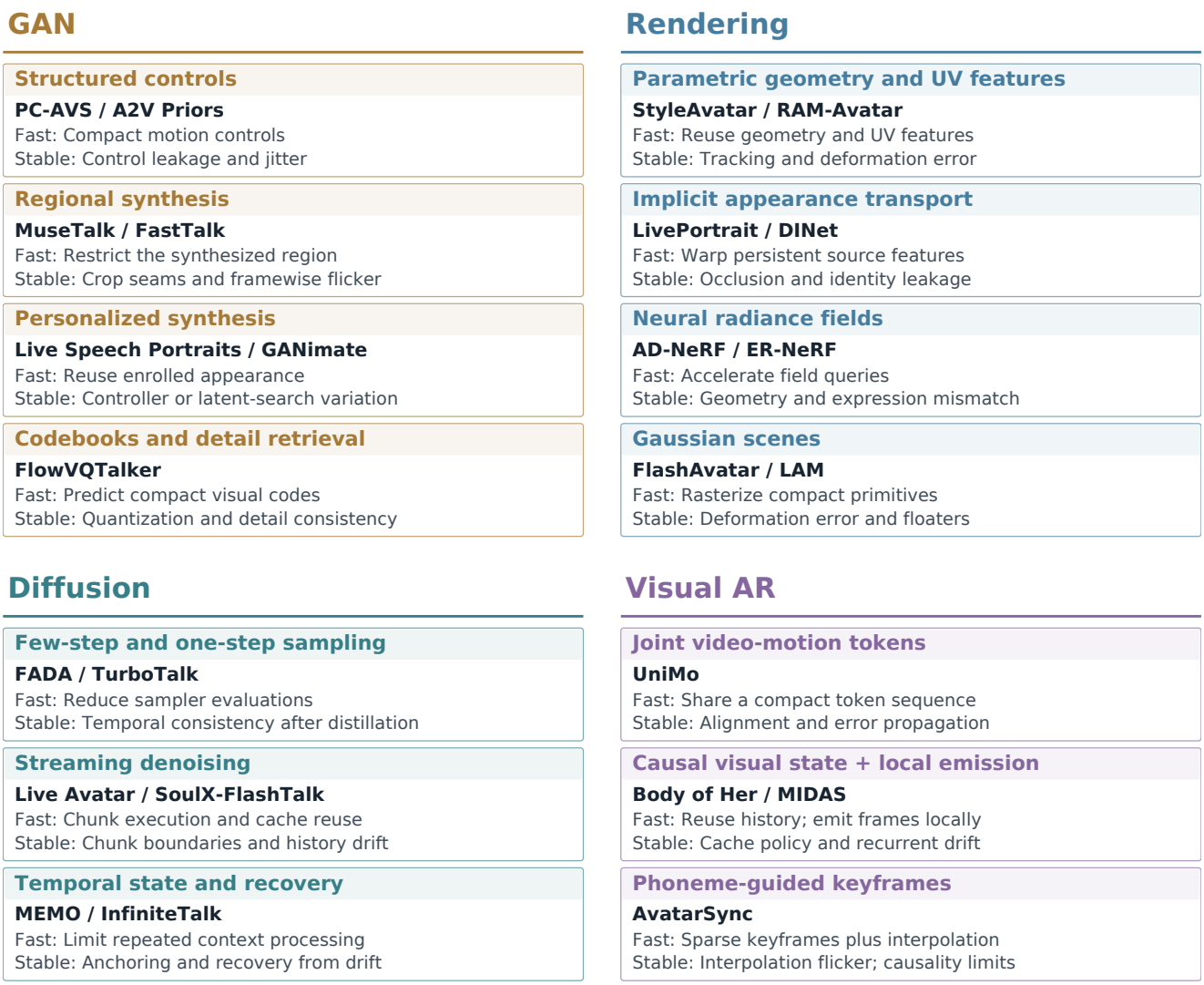


Fig. 4 Representative methods for 14 selected mechanism groups across four visual-owner routes. Figure 3 gives the full overview. Each group highlights a computation lever (Fast) and a temporal-state issue or design target (Stable), based on the analysis in Sections 3–6. These annotations do not certify performance. Mechanisms can overlap; primary-route assignment follows Section 2.3.

2.3 Ordered Route Ownership

Hybrid systems cannot be classified reliably by their motion predictor or training loss alone. We inspect the representation, temporal update path, and RGB reconstruction stage, then assign the first applicable route in this sequence (Figure 4):

1. **Render-based.** A subject-specific explicit or neural scene representation, including a 3D mesh, neural radiance field, 3D Gaussian set, or identity-conditioned spatial feature warp, owns appearance and is geometrically queried or deformed before rendering. Any upstream network that predicts compact motion latents, 3DMM coefficients, deformations, or Gaussian offsets acts as a motion controller rather than the visual owner.
2. **Autoregressive.** If the system is not render-based, a native autoregressive model owns the cross-frame dense or discrete visual state, predicting evolving visual tokens or frame latents from prior visual states. A local diffusion or flow head may refine or decode the current frame's representation, provided the cross-frame causal state is updated independently of the local sampling trajectory.
3. **Diffusion-based.** If neither of the preceding rules holds, a diffusion or flow-matching backbone owns dense pixels, dense visual latents, or sequential visual chunks, regardless of whether a generic variational autoencoder (VAE) decodes the resulting latents to RGB.
4. **GAN-based.** Otherwise, an adversarially trained feed-forward generator owns final dense or regional image synthesis. A generator evaluation is feed-

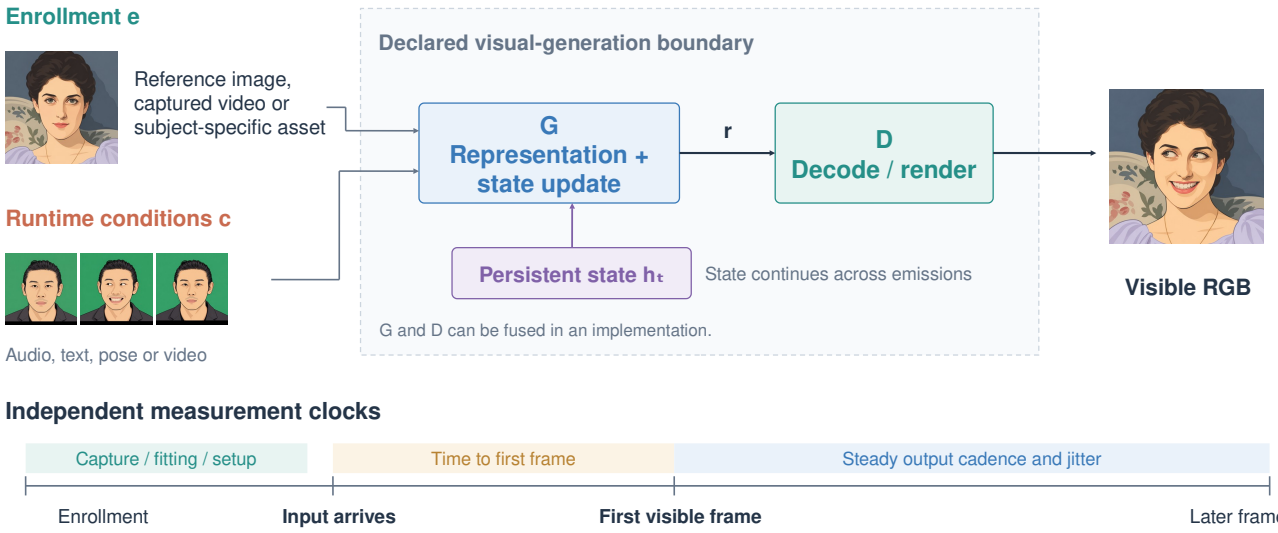


Fig. 5 Generation boundary and measurement intervals. G updates the visual representation and state. D decodes or renders RGB. They may be fused. The clocks distinguish enrollment, time to first frame, and output cadence and jitter. The portraits, architecture and clocks are schematic.

forward, but a complete pipeline may repeat it during test-time latent optimization; this changes execution cost rather than the appearance owner. Attaching an adversarial discriminator solely as an auxiliary training loss on a warper, neural field, or diffusion decoder does not transfer route ownership.

Thus, diffusion motion driving a Gaussian avatar belongs to rendering. An AR conversational planner conditioning video diffusion belongs to diffusion. Secondary modules are recorded as auxiliary roles.

Operational ownership tests for hybrid pipelines. Auxiliary conditioning, state, and decoding modules can govern latency or drift even when they do not own appearance. Figure 6 illustrates two recurring ownership boundaries.

The AR/diffusion boundary depends on where current-frame sampling noise enters the temporal graph. In native visual AR, the backbone computes $h_t = F(c_{\le t}, z_{<t})$ from clean or committed prior states, independently of the current unit's noise. A local diffusion or flow head then samples z_t conditioned on fixed h_t (Figure 6(a)). In block-causal diffusion, the temporal backbone instead processes the changing noisy chunk $z_t^{(s)}$ and noise level s at each denoising step: $v_\theta(z_t^{(s)}, s, z_{<t}, c)$ (Figure 6(b)). Both generate causally across time, but only the first keeps cross-frame state updates independent of local sampling.

The rendering boundary depends on geometric queries or spatial transport of subject-specific appearance before decoding (Figure 6(c)). StyleAvatar's UV features and StyleUNet decoder satisfy this test, as does LivePortrait's keypoint-driven source-volume warp. Both belong to rendering despite their adversarial training. Static feature banks accessed only by concatenation or attention do not satisfy the test.

2.4 Module Roles in Hybrid Systems

Table 2 separates the visual owner from conditioning, temporal state, motion control, and decoding in ten representative systems. This decomposition identifies where latency and errors enter a pipeline. Audio buffering or reference tracking may govern startup, while another module limits throughput or accumulates temporal drift.

2.5 Route-Specific Cost Structures

Each route allocates work differently across enrollment, first-frame delivery, and steady-state generation. Let C denote additive work in a declared unit, such as floating-point operations (FLOPs), GPU-seconds, or serial stage milliseconds. Asynchronous stage times cannot be summed into wall-clock latency. For m ideal stages with durations $t_1, \dots, t_m$, first-output latency is $\sum_i t_i$ plus input buffering. Steady-state chunk cadence is bounded by $\max_i t_i$, with further communication and contention costs. Enrollment cost C_{enroll} is reported separately or amortized over a declared session length T.

GAN-based synthesis. An adversarially trained feed-forward generator maps controls or latents to a frame or facial crop in one network evaluation [64, 97, 98]. In direct one-call pipelines, latency depends on backbone

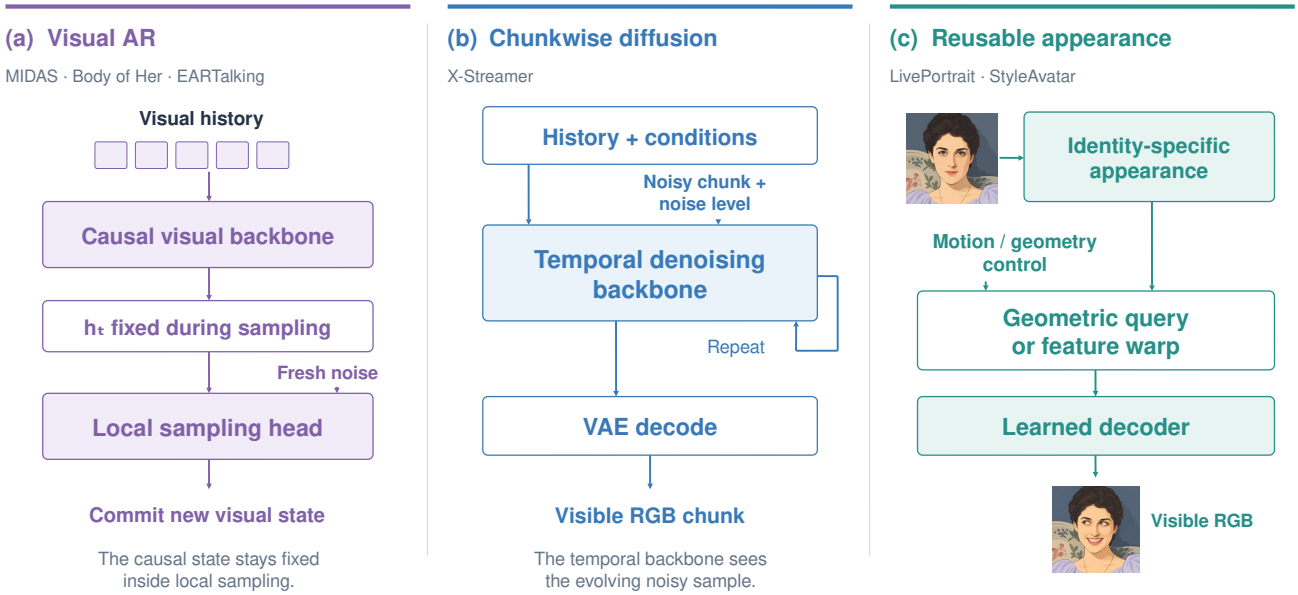


Fig. 6 Visual ownership in hybrid architectures. (a) MIDAS, Body of Her, and EARTalking hold causal visual state fixed during local sampling [5, 28, 225]. (b) X-Streamer's temporal backbone repeatedly denoises the current chunk [229]. (c) LivePortrait and StyleAvatar reuse appearance through feature transport or geometric queries [68, 212]. Architectures and portraits are schematic.

capacity, resolution, input feature extraction, and post-synthesis composition. Systems that optimize latents at test time must additionally count all repeated generator evaluations, as illustrated by GANimate [59].

Diffusion and flow matching. Diffusion models [75] and flow-matching networks [57,127] map noise to visual data through iterative denoising or vector-field integration. Total cost scales with the spatiotemporal token volume of the generated unit, the strict number of function evaluations (NFE), per-step network complexity, multimodal conditioning, and VAE decoding [104, 163, 172]. Step distillation [138,174,189], cross-step feature caching, and multi-GPU pipeline parallelism target different multiplicative factors within this cost product.

Autoregressive visual state. Visual autoregressive models predict discrete tokens [153] or continuous latents [25] sequentially from preceding visual context [15, 169, 231]. Per-frame cost depends on token granularity, serial decoding depth, attention span, key–value (KV) cache maintenance, and local emission or codec reconstruction. While KV caching eliminates redundant historical projections, unbounded context growth increases memory traffic and latency unless controlled by sliding windows, token compression, eviction, or parallel speculative emission [111].

Render-based synthesis. Render-based pipelines invest upfront computation in constructing or encoding a subject-specific appearance asset, then animate that asset per frame using compact driving controls. Runtime work spans control tracking or prediction, spatial deformation, volumetric ray marching or Gaussian splatting, neural shading, and background composition, where NeRF ray integration and 3DGS primitive sorting impose markedly different per-frame costs [99, 149, 152]. Although the persistent scene representation anchors static identity, temporal artifacts can still enter through tracking noise, deformation extrapolation, or seam blending.

2.6 Conditions and Intermediate Representations

Audio, text, pose, expression, gaze, and video specify runtime behavior. Their feature extraction and tracking costs belong to the end-to-end budget [20, 248]. Enrollment supplies appearance through a portrait, monocular clip, calibrated multi-camera recording, or rigged asset.

Intermediate representations determine controllability and cost. Sparse landmarks and keypoints provide geometric guidance without dense texture. Facial action units [56], 3DMMs, FLAME, the Skinned Multi-Person Linear Model (SMPL), SMPL-X, and blendshape rigs expose pose and expression parameters [119, 134, 159]. Dense flow and feature warps transport source appearance, while implicit motion latents trade interpretability for expressive capacity. Visual tokens and spatiotemporal latents carry scene appearance. Neural fields and Gaussians store subject geometry and radiance [99, 149]. Compact controls do not necessarily imply fast RGB output: low-dimensional 3DMM prediction may feed an expensive volumetric renderer, while a lightweight VAE may decode dense latents quickly. We therefore compare

Table 2 Module roles in representative systems. Inputs list driving conditions in addition to the source identity or reference. Portrait output includes a face crop where specified. DC-AE denotes Deep Compression Autoencoder.

Method, task / input	Year	Visual owner, route / supporting component	Temporal state and motion/control owner	Decoder / renderer, compositor
Wav2Lip+ GAN [164], face crop / audio + source video	2020	RGB face generator, GAN / training-only sync expert	No recursive visual state, masked source pose and audio condition each frame	Feed-forward face decoder, crop insertion is a separate output stage
MuseTalk [268], lower face / audio + source	2024	Regional latent predictor, GAN / generic VAE	Framewise visual prediction, audio conditioning and reused blend mask	VAE image decoder, mask-based blending
PersonaLive [123], portrait / driving video	2026	Dense visual diffusion, diffusion / VAE	Selected visual history and reference, motion conditions drive chunk synthesis	VAE variants
Live Avatar [87], portrait / audio	2025	Dense chunk denoiser, diffusion / causal VAE	Reference sink and rolling KV, audio-conditioned denoising	Causal VAE on a separate GPU in full system
MIDAS [28], human video / multimodal controls	2025	Continuous frame tokens, visual AR / local diffusion	AR backbone context, separate codec history, condition tokens	Four-step local head, then DC-AE
EARTalking [225], portrait / audio + frame controls	2026	Frame visual state, visual AR / local denoising	Causal context, reference sink and cyclic positions, frame-level control	Local sampling and 3D-VAE
StyleAvatar [212], portrait / tracked motion	2023	Enrolled facial UV features, rendering / adversarial decode	Fixed subject appearance, 3DMM tracking drives per-frame geometry	OpenGL UV rendering and StyleUNets, learned regional masks
LivePortrait [68], portrait / driving video	2024	Source feature volume, rendering / learned decode	Source identity and keypoint-driven deformation, motion extractor	Feature warping and neural decoder, paste-back stage
DyStream [23], dyadic portrait / audiovisual input	2025	Reused appearance volume, rendering / motion generation	Motion controller with anchor, RGB appearance persists outside controller	Portrait feature warping and rendering
OmniResponse [137], dyadic portrait / audiovisual conversation	2025	Pretrained portrait backend, visual route unspecified / AR response control	Multimodal AR model predicts facial coefficients and text, TempoVoice predicts speech	Coefficient decoder followed by portrait renderer

anatomical scope, personalization, representation owner, decoder, and timing boundary independently.

2.7 Common Acceleration and Measurement Terms

Across all four routes, acceleration techniques reduce runtime cost through five principal levers [178]:

- **Fewer evaluations:** step distillation, consistency training, high-order ordinary differential equation (ODE) solvers, blockwise token prediction, and speculative decoding reduce strict NFE or sequential network calls [111, 138, 189, 251].
- **Cheaper evaluations:** latent compression, sparse or localized attention, lightweight backbones, weight/activation quantization, and hardware-aware kernels lower the cost of each forward pass. In attention layers, input/output (I/O)-aware FlashAttention restructures exact 16-bit floating-point (FP16) or bfloat16 (BF16) computation to minimize high-bandwidth memory traffic [47, 48]. FlashAttention-3 exploits Hopper 8-bit floating-point (FP8) Tensor Cores alongside block quantization and incoherent processing to bound numerical error [176]. How much these kernel speedups reduce end-to-end avatar latency depends on the surrounding non-attention stages.
- **Computation reuse:** denoising-step caches, static reference caches, condition caches, and rolling KV caches reuse intermediate activations or projections across sampling steps, frames, or chunks [128, 140, 143].
- **Decoder and renderer reduction:** pruned or distilled VAEs, bounded ray sampling, Gaussian primitive culling, fast rasterizers, and localized crop synthesis reduce the cost of mapping intermediate states to RGB pixels.
- **System optimization:** graph compilation, mixed precision, memory layout optimization, asynchronous I/O, and multi-device pipeline parallelism reduce scheduling overhead and overlap serial stages.

Throughput measures completed output per unit time. Frames per second (FPS) must specify resolution and pipeline boundary, and differ from the media playback rate. The real-time factor (RTF) is generation time divided by media duration, with RTF < 1 indicating faster-than-playback execution. For N frames generated in w seconds at playback rate f, FPS $= N/w$ and RTF $= wf/N = f/\text{FPS}$ under the same boundary. *First-output latency* starts at input arrival. Time to first frame (TTFF) ends at the first committed decoded RGB frame. For example, a 12-frame chunk generated in 0.30 s yields 40 FPS. Buffering 0.48 s of audio at 25 FPS first raises minimum TTFF to 0.78 s. *Inter-chunk latency* and jitter measure intervals between committed chunks. Strict *NFE* counts denoiser or vector-field calls per generated unit, including guidance branches and cache refreshes.

Real-time execution meets playback deadlines for a declared workload. *Streaming* commits chunks before the input sequence ends. *Causal* generation uses past and current inputs up to a stated look-ahead buffer. *Long-horizon* generation extends well beyond the training clip window. These properties do not imply one another. Runtime comparisons must specify hardware, device count, resolution, precision, concurrency, warm-ups, timer endpoints, and included stages.

2.8 Failure Modes and Stability Mechanisms

Sustained generation must preserve identity and scene coherence while following changing speech, expression, and pose. Five failure modes recur: *appearance drift* in identity, color, or texture, *temporal discontinuities* between frames or chunks, *control failures* such as desynchronization or damped motion, *geometric failures* in tracking, deformation, or seams, and *service failures* such as stalls, memory growth, or resets. These can originate outside the visual owner. Table 3 separates seven diagnostic dimensions to identify their causes.

Five recurring mechanisms address long-horizon degradation in streaming generators, especially chunk-wise diffusion and visual AR (Figure 7):

L1: Persistent identity anchor. Retaining a clean reference portrait, an enrolled 3D/feature asset, or dedicated attention-sink tokens across all generation steps anchors subject appearance against cumulative drift, though overly rigid anchoring can resist legitimate head rotations or scene transitions.

L2: Bounded multiscale temporal memory. A short sliding window preserves local continuity, while a selective or compressed long-term bank retains earlier context. Both need explicit capacity and eviction rules. We use L2 only when both timescales and replacement policies are documented. Examples include SoulX-LiveAct's reference, long-term, and short-term slots and InteractiveAvatar's diversity-promoted bank [188, 281]. Simple first-in-first-out (FIFO) KV windows, codec caches, and PersonaLive's unspecified history selection do not meet this definition [123].

L3: Position re-indexing. Temporal indices are reassigned as the context window advances. Methods use rotary position embedding (RoPE) re-indexing, including pre-RoPE key caching and cyclic or Rolling RoPE schedules. EARTalking and Live Avatar use these policies to limit positional extrapolation during streaming [87, 225].

L4: History-robust training. Training on imperfect history reduces the mismatch between clean training context and inference errors. *L4-G* uses the generator's own rollouts, as in Self-Forcing [86]. *L4-C* corrupts ground-truth history with noise or token dropping, as in MIDAS, EARTalking, and Hallo2 [28, 43, 225].

L5: Explicit boundary-state transfer. Passing a dedicated motion latent, boundary frame representation, or overlapping temporal state across adjacent chunks, such as Motion Condition Injection in JoyStreamer-Flash or motion propagation in Avatar V [114, 124], enforces cross-chunk kinematic continuity beyond standard causal attention.

L1–L5 describe mechanisms whose effects depend on duration and input diversity. EARTalking ablates components over fixed multiples of the training window, while Live Avatar tests multi-hour survival with looped audio under stationary conditions [87, 225]. Section 8.2 defines continuous-session protocols with time-binned metrics.

3 GAN-Based Methods

Direct adversarially trained feed-forward generators produce a frame or facial crop in one network evaluation [64, 97, 98]. This avoids a visual denoising trajectory, but does not rule out repeated evaluations during latent optimization. Feature extraction, resolution, blending, and restoration still govern deployment latency. Regional synthesis, compact motion controls, and personalized appearance reuse reduce recurring work. Framewise jitter and controller drift remain concerns even without recursive visual prediction.

3.1 Scope, Ownership, and Cost Boundary

The GAN route requires an adversarially trained feed-forward network to own dense or regional synthe-

Table 3 Dimensions of temporal stability and the components that can affect them. Interpret measurements together with generation duration, input conditions, and reset behavior. Low variation alone is insufficient if the output has poor quality or fails to follow intended changes.

Dimension	Observable and measurement	Candidate responsible components	Interpretation limit
Identity preservation	Reference-aligned identity similarity over time, failed-detection fraction	Appearance representation, visual generator, identity conditioning	A consistently wrong identity can have little drift, pose changes affect recognition.
Appearance preservation	Region-specific clothing/background/color or detail consistency, artifact ratings	Generator, decoder, appearance asset, compositor	Mask intended lighting, clothing, or scene changes according to the test condition.
Motion continuity	Pose/landmark trajectories, frame/chunk jumps, target error where paired	Motion controller, state update, tracker	Low movement may indicate frozen or suppressed motion.
Audiovisual synchronization	Fixed-window lip-sync measures over time, shift/error distribution	Audio encoder, motion control, audiovisual scheduling	Synchronization does not establish identity or expression fidelity.
Geometry/rendering stability	Wobble, disocclusion, deformation and seam errors in a declared coordinate frame	Tracking, deformation, scene representation, renderer, compositor	Static geometry or a fixed identity asset does not exclude visible artifacts.
State/error accumulation	Error trajectories under continuous rollout, matched state/anchor/history ablations	Visual or controller memory, eviction, boundary transfer, training history	A trend alone cannot identify its cause, bounded memory is a resource result.
Service reliability	Completion/failure rate, stalls, queue growth, resets, recovery time, memory trajectory	Scheduler, cache allocation, decoding and I/O	High average FPS can coexist with missed deadlines or repeated resets.

sis under the precedence rules of Section 2.3. Both training and inference matter. Wav2Lip provides a reconstruction-trained base generator and an adversarial variant, Wav2Lip+GAN [164]. Only the latter enters our GAN comparison. MuseTalk uses a pretrained variational autoencoder (VAE) and U-Net, but predicts regional latents in one call without a noise schedule or iterative sampler [268]. Its reconstruction, synchronization, and adversarial losses therefore place it in this route. An auxiliary discriminator on a warper, neural field, or diffusion decoder alone does not change ownership.

For a single-pass feed-forward pipeline, per-frame runtime work decomposes into four additive terms:

$$C_{\text{frame}} = C_{\text{condition}} + C_{\text{generator}} + C_{\text{decode/compose}} + C_{\text{system}}. \quad (3)$$

Here, $C_{\text{condition}}$ covers audio encoding, landmark tracking, or inversion. $C_{\text{generator}}$ covers pixel or latent prediction. $C_{\text{decode/compose}}$ includes decoding, face parsing, blending, and super-resolution. C_{system} accounts for transfers, scheduling, and media I/O. The following mechanisms target these costs separately. Equation (3) assumes one generator call; iterative latent-search pipelines must sum the generator and optimization costs over all calls.

3.2 Regional Synthesis: Shrinking $C_{\text{generator}}$

Regional synthesis reduces $C_{\text{generator}}$ by regenerating the speech-affected facial region and retaining the surrounding frame. Wav2Lip conditions its decoder on an identity reference, a lower-half-masked pose prior, and speech features [164]. A frozen synchronization expert supplies a training loss without recurrent runtime state. Its base and adversarial variants trade synchronization confidence against sharpness, while low-resolution crop replacement can blur teeth and lip interiors in high-definition footage. Later methods improve detail within this regional design. VideoReTalking separates expression neutralization, lip synthesis, and identity-aware enhancement [30]. Methods such as DINet and IP-LAP instead transport reference appearance and are reviewed under rendering (Section 6.3). Other GAN pipelines retrieve oral textures from identity-matched memory [158, 196], or edit pretrained StyleGAN latents for lip synchronization [66, 67, 102].

MuseTalk further reduces synthesis cost by predicting lower-face VAE latents with a multimodal U-Net in one call, followed by mask-based blending [268]. FastTalk also performs one-step regional latent restoration with adversarial training; initialization from diffusion weights does not introduce a visual denoising trajectory [131]. Independent frame prediction can still cause jitter (Section 3.7). Training-only critics improve articulation without inference overhead. TalkLip replaces Wav2Lip's syn-

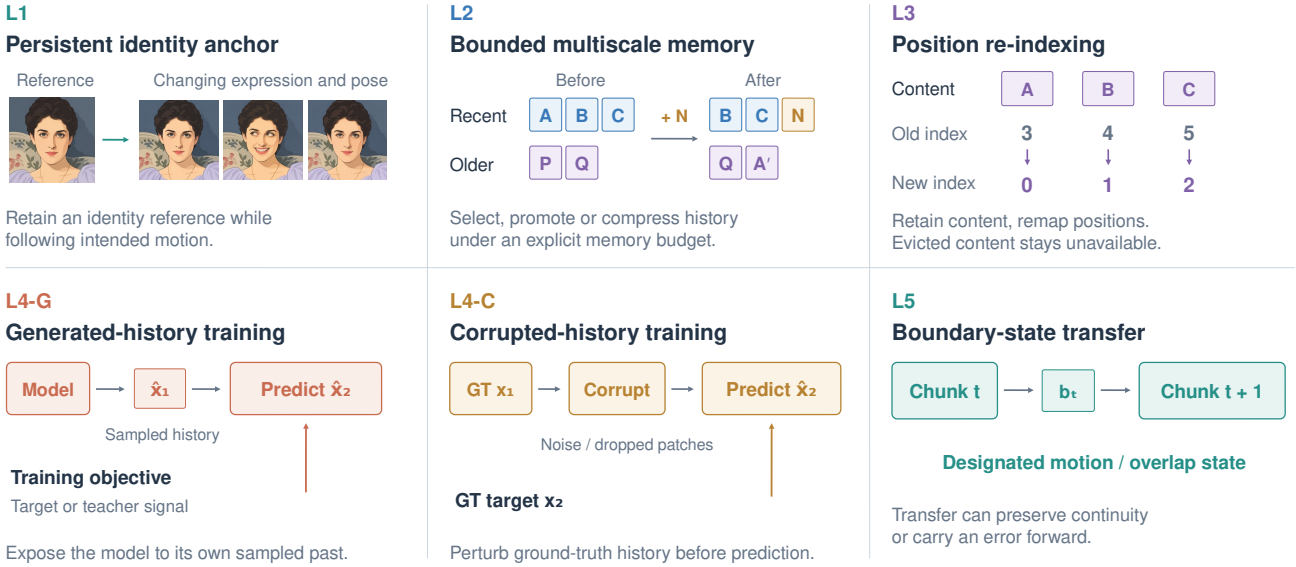


Fig. 7 Operations for sustained generation. L1–L3 retain identity, manage bounded memory, and re-index positions. L4-G and L4-C train with generated or corrupted histories. L5 transfers designated state across chunk boundaries. All panels are schematic. L1 illustrates an identity anchor under changing expression and pose.

chronization loss with a frozen AV-HuBERT lip-reading expert, improving word intelligibility on LRW and LRS2 while keeping the critic off the deployed path [209].

3.3 Structured Control and Codebooks

Full-portrait methods reduce conditioning complexity and constrain variation by separating identity from compact motion controls. MakeItTalk predicts speaker-aware landmarks from audio [288], while PC-AVS separates pose, speech content, and identity in latent space [286]. StyleGAN reenactment methods navigate orthogonal latent directions, mask style channels, or predict hypernetwork modulation [12–14, 22]. Extensions support continuous emotion [7], semantic expression editing [157], cartoon heads [95], and text-guided manipulation [18]. These interfaces compress the driving vector, but inversion and feature extraction remain part of $C_{\text{condition}}$. The full StyleGAN generator also remains on the per-frame path.

Direct adversarial synthesis also supports action-unit, layout, and learned latent controls. FACEGAN transforms action units and pose into landmarks before its face generator and background mixer synthesize RGB [201]. LSR predicts a semantic layout for a SPADE-conditioned image generator [148]. StyleTalker generates compact motion in StyleGAN3's latent space [150]. These controls do not spatially transport a persistent subject appearance field. Audio-to-Visual Diffusion Priors is a useful hybrid: diffusion predicts non-lip motion, but StyleGAN2 owns the final dense RGB synthesis [257].

Codebooks and retrieval memories constrain high-resolution synthesis to learned visual primitives. FlowVQTalker uses autoregressive normalizing-flow controllers for 3DMM expression and pose, then decodes spatial code indices through an adversarial vector-quantized (VQ) generator at 512×512 [195]. Grid-based codebooks provide a related mechanism for person-specific reenactment [211]. FlowVQTalker uses warped-image features as motion guidance before generic VQ-code prediction and image decoding. VideoReTalking likewise uses its expression-normalized face as a structural reference for separate lip synthesis. These hybrid cases are retained as GAN-owned synthesis; the existence of an upstream warp alone is insufficient to establish rendering ownership.

3.4 Personalized Compositional Generators: Amortizing Appearance

Personalized adversarial generators trade enrollment cost (C_{enroll}) for lower recurring synthesis work. Live Speech Portraits projects streaming audio onto low-dimensional manifolds to drive a person-specific generator [135]. StyleAvatar precomputes facial-UV and background features, then runs dynamic StyleUNets and neural-texture composition at runtime [212]. It illustrates static-feature reuse in adversarial decoding, but belongs to rendering because appearance is geometrically queried in UV space (Section 6).

Table 4 Single-H200 runtime (s) and peak memory (GiB) for 8 GAN configurations. R: resident models. Stage, repetition, and attention-backend conventions follow Section 8.3.

Configuration	Year	N	$W \times H$	Attn.	Pre	Gen	Post	Load$_1$	Call $\pm s$	Warm peak M_w (GiB)
Wav2Lip+GAN^R [164]	2020	248	256×256	Eager	0.12	0.09	0.57	1.46	0.78 ± 0.01	3.87
MuseTalk^R [268]	2024	200	704×1216	SDPA$_m$	20.68	0.85	19.99	7.99	41.53 ± 0.06	9.49
TalkLip^R [209]	2023	251	700×700	SDPA$_m$	2.63	0.03	2.22	3.56	4.93 ± 0.02	14.54
Live Speech Portraits^R [135]	2021	672	512×512	Eager	0.99	7.41	10.47	1.92	18.88 ± 0.22	2.08
VideoReTalking^R [30]	2022	147	700×700	Eager	19.01	10.17	31.40	52.65	60.60 ± 0.16	16.01
MakeItTalk [288]	2020	287	256×256	Eager	2.72	2.46	2.39	4.05	10.00 ± 1.25	2.79
PC-AVS^R [286]	2021	361	224×224	Eager	1.34	2.44	11.94	1.97	15.75 ± 0.15	6.64
HyperReenact^R [12]	2023	101	256×256	Eager	7.15	9.12	0.18	16.11	16.48 ± 0.21	5.96

Finding Directions (FD, also referred to as LOR) learns pose and expression directions in StyleGAN2's latent space and uses source-specific pivotal tuning [11]. Tuning belongs to enrollment rather than being hidden in per-frame throughput. GANimate also uses a StyleGAN image prior, but performs iterative latent optimization for each frame [59]. Its reported operating point is about 2 FPS at 1024×1024 on an RTX 2080 Ti; it is not an end-to-end single-pass or established real-time pipeline. These execution boundaries must be reported separately from the GAN route label and from the appearance-reuse mechanism highlighted in Figure 4.

3.5 Open-Source Measurements on One H200

Table 4 reports eight single-H200 GAN configurations under Section 8.3's protocol. Seven retain model objects across calls, while MakeItTalk reloads weights. Surrounding stages often exceed generator cost even in one-call pipelines. MuseTalk generates 200 latent frames in 0.85 s, but preprocessing takes 20.68 s and blending/encoding takes 19.99 s. Total Call is 41.53 ± 0.06 s. PC-AVS spends 11.94 s in restoration and postprocessing versus 2.44 s in Gen. These profiles locate the remaining costs in conditioning and composition once single-pass synthesis becomes inexpensive.

3.6 Fast Design Guidance

Across representative GAN-based pipelines, four recurring acceleration levers map to the cost terms in Equation (3):

1. **Regional synthesis** ($C_{\text{generator}}\downarrow$). Restricting prediction to a mouth crop or lower-face VAE latent reduces generator FLOPs, shifting the evaluation focus to codec reconstruction, blending seams, and oral interior sharpness.
2. **Compact controls** ($C_{\text{condition}}\downarrow$). Orthogonal basis banks, 3DMM coefficients, StyleGAN directions, and VQ codebook indices streamline conditioning at the cost of bounding expressiveness to the chosen control subspace.
3. **Appearance reuse** ($C_{\text{generator}}\downarrow$). Precomputed subject features amortize synthesis over repeated sessions, whereas one-shot detail retrieval preserves identity generality with added feature-alignment overhead.
4. **Fixed-path optimization** ($C_{\text{system}}\downarrow$). Graph compilation, FP16 execution, and fused engines reduce framework overhead once the feed-forward architecture is frozen. Render-owned StyleAvatar provides a transferable TensorRT implementation example, not a GAN ownership example.

3.7 Stable Design Guidance

Because most feed-forward GAN generators synthesize each frame without feeding the output image back as input, they avoid recursive visual error accumulation. However, stateless framewise prediction does not guarantee temporal smoothness: MuseTalk documents single-frame flicker, and noisy per-frame pose estimates or bounding-box detections directly induce spatial jitter. Where recurrent motion controllers are present, as in Live Speech Portraits and FlowVQTalker, state errors can still accumulate along the control path. In terms of the stability taxonomy in Section 2.8, static reference portraits and personalized feature banks act as L1 identity anchors. FlowVQTalker's coefficient history does not constitute bounded multiscale temporal memory (L2). History-robust training (L4) or cross-chunk state transfer (L5) remain largely unexplored in this route.

3.8 Synthesis and Open Questions

Direct one-call adversarial pipelines achieve low core synthesis cost through feed-forward evaluation, localized mouth cropping, compact motion bases, and static

Takeaways: GAN-based generation

Low per-frame latency. Regional, one-call generators reduce synthesis cost. Check crop seams and mouth interiors, and profile preparation: it can dominate the complete call.

High-resolution throughput. Compact controls and retrieval reduce per-frame work but restrict expressiveness. Compiled FP16 serving is a possible extension whose benefit depends on the target hardware.

Repeated subject use. Person-specific models reuse learned appearance across sessions, at the expense of hours of enrollment and no zero-shot support.

Longer sessions. Reference conditioning and temporal controls can constrain variation, but framewise jitter and controller drift remain. Joint long-session evaluation is still missing in the reviewed systems.

feature reuse. However, our stage-resolved timings confirm that fast core prediction does not automatically yield low system latency unless face tracking, feature preparation, and canvas blending are co-optimized. On the stability side, avoiding recursive visual feedback removes one source of progressive texture degradation, yet leaves framewise flicker, crop-boundary seams, and recurrent controller drift unresolved. Establishing how GAN-based editors behave over multi-minute interactive sessions requires time-binned benchmarks that evaluate boundary artifacts, synchronization, and temporal smoothness jointly.

4 Diffusion-Based Methods

Diffusion and flow-matching models synthesize high-fidelity human video conditioned on identity references, speech, text, skeletal pose, and historical context [75, 79, 126, 127, 200]. Overcoming their iterative sampling cost requires compressing the spatiotemporal representation, distilling the trajectory into a few function evaluations, reusing intermediate computations, and pipelining execution across devices. When deployed in streaming sessions, chunkwise generation couples speed directly to stability: each committed chunk becomes conditioning history for subsequent blocks, requiring explicit anchoring and history-robust training to prevent cumulative drift.

4.1 Scope and Cost Model

Denoising diffusion probabilistic models (DDPMs) reverse a discrete noise chain [75]. Flow matching learns a continuous velocity field without simulating ordinary differential equations (ODEs) during training, but generally integrates that field over multiple inference steps [57, 127]. Normalizing flows instead use invertible mappings with tractable Jacobians [156]. Simulation-free training of rectified or optimal-transport flows therefore does not imply single-step sampling. In video diffusion Transformers (DiTs), each step depends on spatiotemporal token count, network depth, and attention span [160, 247]:

$$C_{\text{gen}} = N_{\text{FE}} C_{\text{denoise}} + C_{\text{condition}} + C_{\text{decode/render}} + C_{\text{system}}. \tag{4}$$

Here, N_{FE} counts every denoiser or vector-field call per generated unit, including multi-stage solvers, classifier-free guidance (CFG) branches, and clean-cache refreshes. $C_{\text{condition}}$ covers input encoders, while $C_{\text{decode/render}}$ covers VAE decoding or super-resolution. C_{system} includes scheduling, communication, buffering, and media I/O. As distillation reduces NFE, decoding, conditioning, or synchronization can become the bottleneck. Component speedups therefore do not multiply directly into system gains. TTFF includes input look-ahead, conditioning, first-chunk denoising, and decoding. Subsequent playback depends on inter-chunk latency and jitter, which require separate measurement.

FADA, RAP, StreamAvatar, SoulX-FlashTalk, and Avatar V use noise-conditioned backbones to generate dense visual data and belong to diffusion [55, 124, 179, 192, 284]. IF-MDM, INFP, AudioRTA, Ki et al.'s Avatar Forcing, EmbodiedHead, FloodDiffusion, and OmniTalker instead diffuse motion or 3DMM parameters before querying a subject-specific renderer [17, 101, 109, 223, 244, 269, 292]. They belong to rendering under Section 2.3's rules. Motar remains in diffusion because its downstream decoder performs dense visual diffusion [3] (Figure 6).

Human video has a useful temporal asymmetry: appearance and background often change slowly, while articulation and expression change rapidly. Reference anchors and caches can retain identity, clothing, and layout, but mouth shapes must follow high-frequency audio cues. READ exploits this through aligned speech/video compression [207], RAP through hybrid audio attention [55], and SyncCache through selective residual reuse [139]. Figure 8 shows the execution path and call accounting. State operations follow L1–L5 in Figure 7.

Conditioned generation: historical background. Early diffusion generators established reference, audio, and motion conditioning for human video. DiffTalk, V-Express, Hallo, and EchoMimic combine speech with appearance or landmark guidance [29, 180, 206, 232]; AniPortrait

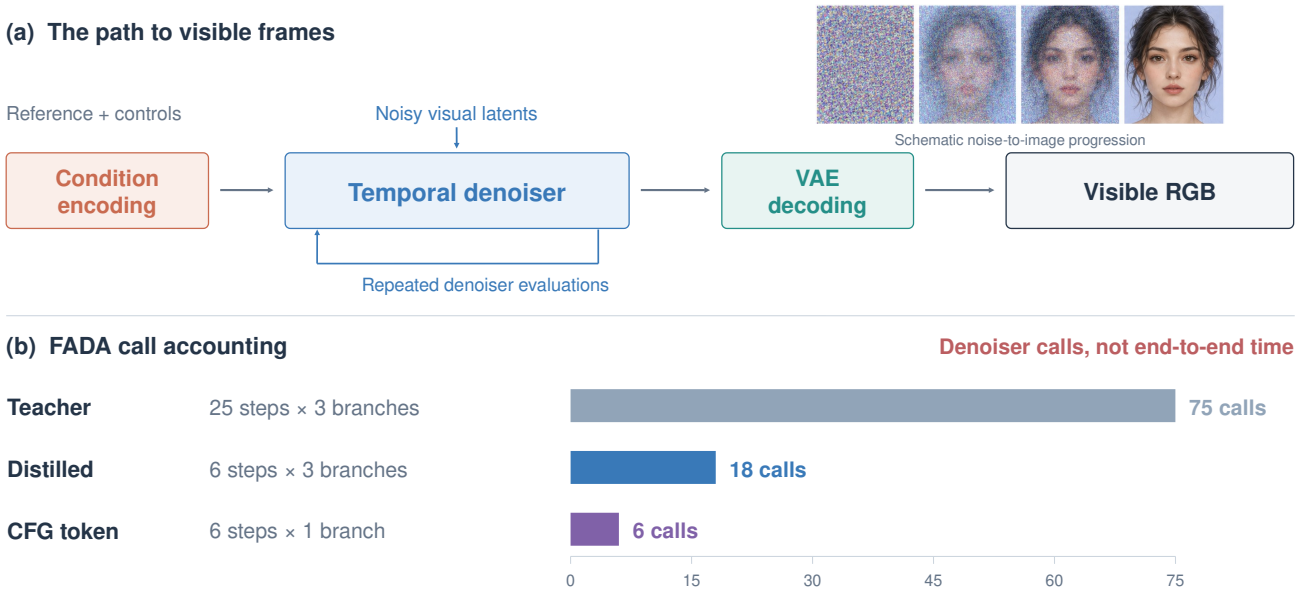


Fig. 8 Diffusion execution and call accounting. (a) Schematic path from conditions and noisy latents through repeated denoising to decoded RGB. The noise-to-image progression is illustrative. (b) FADA's reported step and guidance configurations yield 75, 18, and 6 denoiser calls [284]. These counts exclude decoding and do not measure wall-clock speedup.

and Hallo3 extend the conditioning pipeline and visual backbone [44, 224]. DreamPose, DisCo, MagicAnimate, and X-Portrait support pose- or video-driven animation [96, 218, 230, 238]. Diff2Lip and LatentSync target regional synchronization [115, 151], while FantasyTalking, OmniAvatar, and MultiTalk broaden expressive human and multi-person generation [61, 107, 214].

These conditioning and synthesis foundations provide background for the acceleration and state-management mechanisms below. Some already extend video duration through continuation, including EMO and MultiTalk [107, 200]. Longer output alone does not establish real-time operation or stable long-session deployment; these require separate timing and drift measurements.

4.2 Fast I: Compact Representations and Efficient Backbones

Compression reduces per-step cost through two strategies. Motion-space methods diffuse compact controls and delegate RGB synthesis downstream. Examples include IF-MDM and INFP's implicit motion, AudioRTA's expression latents, Ki et al.'s reactive Avatar Forcing, EmbodiedHead's FLAME parameters, and FloodDiffusion's skeletal trajectories [17, 101, 109, 244, 269, 292]. Their renderers are reviewed in Section 6. Motar instead uses a four-step block-causal visual diffusion decoder [3]. Dense visual methods retain appearance synthesis but compress the video tensor with a spatiotemporal VAE before DiT processing. This follows architectures such as Stable Video Diffusion, HunyuanVideo, Wan, CogVideoX, and LTX-Video [10, 71, 106, 204, 247].

For temporal and spatial compression factors r_t and r_s, a T-frame clip at $H \times W$ yields a token count proportional to $THW/(r_t r_s^2)$ [71, 160]. Compression lowers attention work, KV memory, and communication. It can also merge consecutive phonemes and reduce capacity for teeth, lips, and eyelids. Avatar backbones therefore co-design tokenizers and conditioning. READ pairs a Temporal VAE with an aligned SpeechAE [207]. RAP combines LTX-VAE compression, hybrid audio attention, and cross-frame latent inheritance [55]. Xu et al. inject multi-scale reference features into a deep-compression VAE decoder to recover identity detail [235]. With reference videos, attention can itself become a quadratic bottleneck. Avatar V combines asymmetric sparse attention, first-step reference KV/RoPE caching, and one-step super-resolution to reduce this cost [124].

As DiT cost falls, VAE decoding takes a larger share of chunk latency. TinyVAE increases throughput in PersonaLive and MotionStream, but PersonaLive's ablation also shows reduced identity similarity and temporal smoothness [123, 181]. Wan supports streaming reconstruction by caching adjacent features inside a chunked causal VAE [204].

Conditioning cost and expression control. Expressive conditioning adds costs that belong in $C_{\text{condition}}$. EmotiveTalk separates phonetic cues from emotional features and expands context through emotional video dif-

fusion [208]. HunyuanPortrait uses attention adapters to inject identity-disentangled motion features from a reference video [237]. Their feature extraction and cross-attention should be profiled alongside synchronization, naturalness, and identity preservation.

4.3 Fast II: Few-Step and One-Step Distillation

Few-step sampling uses three distinct mechanisms. Rectification or consistency training simplifies mappings across noise levels [138,189]. Distribution matching distillation (DMD/DMD2) aligns student distributions using real and teacher score gradients, two-timescale updates, adversarial losses, and backward simulation [250, 251]. CFG distillation folds guidance branches into one pass. FADA illustrates why solver steps and strict NFE differ (Figure 8(b)). Its 25-step, three-branch denoising diffusion implicit model (DDIM) teacher uses 75 denoiser calls (NFE-D). Six-step PeRFlow distillation reduces this to 18, and a learned guidance token folds the branches to reach 6 [284]. Cache maintenance adds calls in streaming. Live Avatar's four-step DMD removes CFG, but standard Self-Forcing still needs a clean-latent pass to refresh history, totaling 5 NFE. Live Avatar's History Corrupt instead retains step-matched noisy KV, yielding 4 NFE and enabling timestep pipelining [87].

The generated unit and trajectory also affect cost. LLIA combines two- or four-step consistency/adversarial distillation, 8-bit integer (INT8) TensorRT execution, and overlapping U-Net/VAE stages [254]. TurboTalk distills a four-step DMD model to one evaluation per chunk [130]. In Cui et al.'s sliding-window AvatarForcing, one joint update per shift still gives each block four updates before emission [46]. LeapTalk instead uses a reference-anchored Brownian bridge with signal-to-noise ratio (SNR)-aligned time warping [265].

Training only on ground-truth context leaves few-step students unprepared for their own rollout errors. MotionStream, StreamAvatar, and LiveTalk use on-policy rollouts (L4-G) [31, 181, 192]. SoulX-FlashTalk uses multi-block retrospective self-correction [179]. AsymTalker uses asymmetric teacher/student history conditioning [136]. These designs expose the student to its own synthesized history during training. Generic distribution matching can still blur articulation or damp motion. Lip Forcing restricts teacher CFG to intermediate noise and weights DMD gradients by SyncNet confidence [33]. Hallo-Live adds audiovisual preference rewards [113]. DynaForcing uses data-anchored rollouts, rewards, and reference perturbation to counter motion damping [88]. Routed Forcing supervises non-mouth regions with real data at high noise while retaining distribution matching for facial detail [191]. Evaluation must therefore pair NFE and FPS with motion diversity, synchronization, and multi-minute drift.

4.4 Fast III: Computation Reuse and System Parallelism

Cross-step reuse lowers average denoising cost when multiple evaluations remain. LightningCP caches high-level decoder features at non-key timesteps and concentrates fine attention on the foreground [133]. Follow-Your-Emoji-Faster uses timestep-aware caching across 30 DDIM steps [142]. Related approaches cache velocity-field segments [143] or use speculative draft-and-verify decoding [82,111]. SyncCache separates slow appearance from fast speech: early blocks monitor portrait-weighted feature differences, heavy DiT residuals are reused, and lightweight audio adapters run at every step. Accumulated error triggers a full refresh [139].

Denoising-time caches reuse activations within one chunk. Sequence-time caches retain KV or latent history across chunks and directly affect long-horizon state. Avatar-Forever's ForeverCache resets at each chunk boundary, so a separate recovery adapter addresses rollout drift [118]. Vidu S1's TwinCache selects noisy or clean historical KV according to denoising stage [263]. Avatar V caches step-invariant reference projections. SoulX-LiveAct uses Neighbor Forcing to align adjacent chunks at matched noise levels and ConvKV to compress older history [124, 281]. Hybrid KV compression extends this approach [92].

Scheduling determines how call and cache reductions translate into streaming latency. LLIA and StreamAvatar overlap denoising with VAE decoding on one or two GPUs [192, 254]. FlowAct-R1 combines three-NFE sampling, selective FP8, hybrid sequence parallelism, fused kernels, and asynchronous decoding [213]. SoulX-FlashTalk pipelines audio encoding, four-step denoising, VAE decoding, and history encoding across GPUs [179]. Live Avatar assigns its four denoising steps to separate GPUs with step-matched rolling KV, then places the causal VAE on another device [87]. Section 7 analyzes the throughput/startup trade-off.

4.5 Open-Source Measurements on One H200

Table 5 reports our single-H200 measurements across 16 open-source diffusion configurations, including sliding-window AvatarForcing, Moore-AnimateAnyone, and SoulX-LiveAct [46, 79, 281], where Gen measures core denoising and Post captures VAE decoding and video export (Section 8.3). Because all 16 released scripts reload model weights on every invocation, model initialization

Table 5 Single-H200 runtime (s) and peak memory (GiB) for 16 diffusion configurations. Stage, repetition, and attention-backend conventions follow Section 8.3.

Configuration	Year	N	$W \times H$	Attn.	Pre	Gen	Post	Load_1	Call $\pm s$	Warm peak M_w (GiB)
StableAvatar [202]	2025	97	512×512	$\text{FA2+SDPA}_{f,e}$	2.30	138.55	3.30	43.04	179.37 ± 2.66	29.30
InfiniteTalk [243]	2025	81	896×448	FA2+xF+SDPA_{e}	13.83	678.05	12.98	224.63	916.74 ± 30.07	45.28
LiveTalk [31]	2025	78	512×512	FA2+SDPA_{e}	0.15	2.11	2.17	110.68	94.89 ± 26.13	21.06
AvatarForcing-SW [46]	2026	129	624×624	FA2+SDPA_{e}	0.51	6.40	6.33	112.07	111.51 ± 20.67	23.76
EverAnimate [121]	2026	77	832×480	FA2+SDPA_{e}	159.26	104.53	5.54	51.65	292.92 ± 5.78	57.28
LeapTalk [265]	2026	28	512×512	FA2+SDPA_{e}	0.20	0.13	0.72	6.01	4.49 ± 0.19	9.81
Avatar-Forever [118]	2026	33	768×512	SDPA_{e}	7.14	1.64	1.20	18.25	17.90 ± 0.26	41.82
TaoMate [62]	2026	121	768×512	SDPA_{e}	19.88	6.22	3.28	47.12	48.34 ± 2.33	65.50
Moore-AnimateAnyone [79]	2024	16	512×784	SDPA_{m}	0.63	17.65	1.73	34.07	43.28 ± 4.65	14.93
SoulX-LiveAct [281]	2026	245	512×512	SDPA^{S}	10.99	20.92	7.13	99.37	123.24 ± 13.39	76.18
Follow-Your-Emoji-Faster [142]	2026	16	512×512	xF+Eager^{S}	2.44	3.72	0.36	37.18	40.86 ± 3.24	21.55
Hallo-Live [113]	2026	21	992×512	FA2+SDPA^{S}	0.03	1.02	1.12	13.70	13.77 ± 0.04	49.29
LightningCP [133]	2026	16	512×512	SDPA^{S}	4.17	4.57	0.59	15.81	22.68 ± 0.08	40.64
Lip Forcing [33]	2026	9	512×512	SDPA^{S}	3.56	1.21	0.58	243.82	202.42 ± 35.75	50.31
Live Avatar [87]	2025	45	704×384	$\text{SDPA}_{c}/\text{FA3}^{S}$	17.23	7.22	8.04	79.12	98.58 ± 0.25	74.75
PersonaLive [123]	2026	16	512×512	xF+Eager^{S}	0.63	1.06	1.04	23.68	18.13 ± 0.27	15.82

dominates single-call elapsed time in distilled pipelines: LiveTalk requires 110.68 s for first-call loading (Load_1) and averages 94.89 s per warmed Call despite spending only 2.11 s in Gen. Outside the denoiser, conditioning preprocessing (Pre) takes 159.26 s in EverAnimate and 19.88 s in TaoMate, underscoring why end-to-end serving requires persistent model residency and optimized conditioning alongside few-step sampling.

4.6 Streaming Generation: Where Fast Meets Stable

Streaming couples chunk size to responsiveness and error propagation. Short chunks reduce buffering and TTFF but require more frequent scheduling and decoding. Each emitted chunk also conditions later output. Diffusion Forcing assigns independent noise levels per token for flexible causal conditioning [24]. Self-Forcing trains on on-policy rollouts to expose students to generated-history errors [86]. CausVid, Causal Forcing, and Rolling Forcing use asymmetric distribution matching, architecture-aligned teachers, or sliding denoising windows with rolling KV [129, 252, 290]. Wan-Streamer scales this design to high resolution [83, 85], while Vidu S1 combines DMD with stage-dependent noisy/clean history reuse [263].

Virtual-human streaming combines these causal designs with persistent identity anchors. PersonaLive advances micro-chunks through progressive noise levels [123]. StreamAvatar, LiveTalk, and EchoTorrent pair attention sinks with self-generated or tail-only forced history [31, 146, 192]. SoulX-FlashTalk, SoulX-FlashHead, SoulX-LiveAct, Lip Forcing, and FlowAct-R1 combine self-correction, anchoring, multiscale memory, synchronization rewards, or action planning [33, 179, 213, 256, 281]. JoyStreamer-Flash explicitly transfers motion conditions across chunks [114]. Offline StableAvatar and InfiniteTalk extend duration with overlapping windows without targeting interactive latency [202, 243]. Vidu S2 and FlowAct-R2 allow intentional identity or prompt switches [89, 262]. Their evaluation must distinguish responsive transitions from stale-reference retention. Speech-encoder receptive fields and input buffering also impose a TTFF floor [192, 280].

4.7 Stable: Streaming State Management

Temporal conditioning predates recent streaming systems. Diffused Heads conditions each denoised frame on earlier observations, while Loopy and MEMO extend temporal or memory guidance [94, 190, 282]. Continuation does not guarantee sustained quality: Diffused Heads reports degradation beyond approximately 8–9 seconds. These remain diffusion pipelines because a denoiser processes each current noisy visual unit.

Sustaining generation over extended horizons requires coordinating the L1–L5 mechanisms of Section 2.8 so that bounded memory neither accumulates corrupted textures nor freezes natural motion. Early long-horizon portrait generators such as Hallo2 mitigate autoregressive degradation by applying patch-drop augmentation and Gaussian noise corruption to conditioning history frames during training (L4-C) [43]. In general long-video diffusion, Self-Forcing++ instead trains on short segments sampled from extended self-generated rollouts (L4-G) [45]. LongLive combines a short local window, persistent attention sinks (L1), prompt-triggered KV re-

caching, and streaming long tuning so that mid-stream prompt switches discard stale semantic context while preserving visual continuity [242].

Structured memory retains longer context under bounded capacity. InteractiveAvatar promotes diverse tokens from a short FIFO window into a fixed-size long-term bank [188]. TaoMate separates active context from bounded audiovisual memory and gates appearance updates against reference anchors [62]. LiveAnimate combines a permanent sink, pose-matched retrieval, a rolling window, and clean-KV refreshes [270]. Live Avatar pairs an Adaptive Attention Sink (L1), Rolling RoPE (L3), and step-matched noisy KV [87]. Its long rollouts under stationary priors motivate further testing with non-repeating interactive inputs.

Restoration and anchoring also address texture decay and pose changes. Gloria retains scene, viewpoint, and expression anchors [246]. EverAnimate combines cross-clip latent propagation with restorative flow matching [121], while TT-SAC re-encodes initial outputs for a training-free refinement pass [274]. LPM 1.0 coordinates speaking, listening, idling, and reflecting through pre-RoPE KV [258]. FlowAct-R1 couples structured memory with generated-history replacement for full-body behavior [213]. These designs, like ARIG's rendering-route summary cache [70], need time-binned drift evaluation beyond memory bounds and qualitative survival demonstrations.

4.8 Design Guidance

As NFE falls, re-profile the full pipeline to identify costs shifted to decoding or conditioning. Pair stage overlap for throughput with small initial chunks for lower TTFF. Evaluate cache compression over long rollouts to detect motion damping or texture drift.

4.9 Synthesis and Open Questions

Dense diffusion and flow-matching pipelines have progressed from slow offline synthesis to real-time multi-GPU streaming by combining spatiotemporal token compression, few-step distribution matching, selective feature caching, and on-policy history training. Despite these advances, two fundamental tensions remain open. First, aggressive latent compression and step distillation can erode fine oral articulation and damp full-body dynamics unless counteracted by targeted rewards or region-routed supervision. Second, attention sinks (L1), multiscale caches (L2), Rolling RoPE (L3), and self-forced rollouts (L4-G) curb drift under stationary conditions. Recovery from corrupted states and adaptation to intended reference or scene changes remain open. Early gating policies such as TaoMate [62] need standardized, matched-compute evaluation. When a causal backbone instead predicts clean or committed visual states directly and relegates diffusion to a local per-frame emission head, the architecture transitions to the native visual autoregressive route examined next.

Takeaways: Diffusion-based generation

Faster fixed clips. Step caches, CFG folding, and latent compression reduce computation. Check stale features near detail steps and compression errors around mouths and teeth.

Playback-rate streaming. Few-step DMD, rolling KV, DiT/VAE overlap, FP8, and fused kernels address different costs. Account for multiple GPUs, TinyVAE quality loss, and generated-history training cost.

Lower startup latency. Smaller initial chunks and staged startup reduce time to first output. Input buffering and startup quality still constrain the gain.

Minute-scale stability. Anchors, generated-history training, and position management address drift. Validate the memory policy, static-scene bias, and loss of motion diversity together.

Hour-scale operation. Bounded memory, anchors, and position management support long rollouts. Adaptive refresh and recovery remain directions to test, including output between sampled windows and under varied inputs.

5 Autoregressive Methods

Visual autoregressive (AR) models predict evolving cross-frame visual tokens or dense latents from preceding visual context, as in MIDAS and Body of Her [5, 28]. Models that predict only text, speech, pose, or facial coefficients act as controllers when another module owns appearance. Likewise, a sequential chunk scheduler does not change a dense denoiser's diffusion ownership.

Autoregressive modeling develops along two independent axes. *State ownership* extends from dialogue and motion planning to decodable visual representations. *Interaction scope* extends from one-way animation to dyadic listening, full-duplex conversation, and persistent world grounding. This distinction allows interactive systems, including those described as world models, to inform causal design while retaining their renderer or diffusion visual ownership. Within native visual AR, cached key-value (KV) states avoid redundant historical projections. Expanding context also increases memory traffic and can propagate early errors. Token compression, quantization, parallel local emission, and speculative decoding address these costs [82, 111].

5.1 Scope, State Ownership, and Cost Boundary

Under Section 2.3's rules, subject-specific meshes, neural fields, Gaussian scenes, and feature warps take rendering precedence. Otherwise, a system belongs to visual AR when its backbone predicts the next frame's or block's visual state from previous visual states. A local diffusion or flow head may then refine the current representation (Figure 6(a)). Runtime work for an emitted frame or block at step t is

$$C_t = C_{\text{condition}} + C_{\text{AR}}(h_t) + C_{\text{local-emission}} + C_{\text{decode/render}} + C_{\text{system}}, \tag{5}$$

where h_t is the persistent visual state at step t. $C_{\text{AR}}(h_t)$ includes causal attention, token prediction, and KV updates. $C_{\text{local-emission}}$ includes denoising or flow steps within the continuous emission head. Other terms cover conditioning, codec reconstruction, data movement, and media I/O. Thus, one AR update per frame can still require multiple local sampling calls and a separate codec pass on the critical path.

5.2 Mechanism Map: Compression, State, and Emission

Designing a visual AR generator requires balancing three coupled architectural choices [5, 28, 231]:

1. **Token granularity and spatial compression.** A frame may contain hundreds of discrete VQ tokens [153], a small set of continuous tokens [25], or one frame latent. This choice determines the backbone's serial sequence length. Compression and parallel within-frame prediction reduce serial depth, while shifting work and fine-detail reconstruction to the tokenizer and decoder.
2. **Persistent state management.** KV caching avoids repeated projections of historical frames, but unbounded context increases memory traffic and retains corrupted tokens. Windows, summaries, eviction, and resets bound $C_{\text{AR}}(h_t)$ but may discard initial identity cues or conversational context. These policies should be distinguished from feature caching inside the downstream codec.
3. **Local emission head.** Compact diffusion or flow heads sample continuous visual tokens conditioned on the AR state, avoiding discrete-codebook quantization artifacts [5, 28]. Their strict NFE and the subsequent codec reconstruction both contribute to frame latency.

5.3 Autoregression at Different Pipeline Levels

Whether an autoregressive module belongs to the visual AR route or acts as an auxiliary controller depends on the variable it predicts. Teller, ARIG, DyStream, and Talker-T2AV predict compact motion or behavioral controls for separate appearance warpers [23, 70, 249, 280]; UniLS supplies facial and body motion [38]. UniTalker instead couples AR speech and landmark prediction to an EchoMimic-based iterative image generator, so its primary visual owner is diffusion [81]. By contrast, UniMo interleaves 2D video and quantized 3D motion tokens for general human video–motion modeling, while MIDAS, Body of Her, and EARTalking maintain explicit cross-frame visual state for conditioned digital humans [5, 28, 155, 225].

5.3.1 OmniResponse: Autoregressive Conversational Control

OmniResponse illustrates multimodal AR control of online dyadic behavior with a separate portrait backend [137]. Its backbone conditions on synchronized speaker/listener histories. Chrono-Text Markup represents pauses and word durations alongside text, while TempoVoice predicts aligned speech tokens. A motion decoder produces facial and pose coefficients for a pretrained portrait renderer. Temporal markup and audio alignment support lip/speech synchronization, but the recurrent state contains behavioral controls rather than dense appearance. Evaluation must still include downstream image-generation cost and long-session visual drift. The available backend description does not establish a particular visual-owner route, so we do not infer one.

5.4 Representative Visual-State Autoregression

Five representative systems cover native visual-state prediction and sparse visual-keyframe generation: UniMo bridges general human video and 3D motion [155], whereas MIDAS, Body of Her, and EARTalking target multimodal or audio-driven digital humans. AvatarSync, renamed FluentAvatar in a later version, generates phoneme-guided visual-token keyframes [52]; the figures count this work once using its cited 2025 edition.

MIDAS encodes each frame into at most 60 continuous tokens with a Deep Compression Autoencoder (DC-AE) [25, 28]. A causal backbone updates visual state across blocks. A four-step local diffusion head realizes the tokens before DC-AE decoding. Blockwise conditioning permits short look-ahead, with an eight-block AR context and separate five-frame codec history.

Body of Her also separates cross-frame state from local token sampling [5]. Its causal Transformer takes audio-visual history, text or trajectory prompts, and identity. A compact diffusion head then samples all tokens of the next frame in parallel before decoding. Our visual-AR label applies to this video-generating configuration, not its separate 3D-motion branch. EARTalking maintains framewise state in 3D-VAE latent space [225]. Its GPT-style backbone uses a ten-latent-frame KV window, a persistent sink frame, and cyclic positions, followed by spatial AR and denoising. Comparisons with render-owned DyStream [23] must include state prediction, local emission, and pixel decoding under aligned timing boundaries.

AvatarSync tokenizes the reference image and predicts visual keyframes from phoneme-conditioned context, then uses timestamp-aware interpolation before RGB decoding [52]. The tokens carry image content rather than only motion coefficients. Its keyframe masking and parallel-prediction descriptions do not by themselves establish fully causal cross-keyframe streaming; preprocessing and interpolation remain part of the latency boundary.

Measured coverage and full-duplex interfaces. No native visual-AR system provided a complete open-source inference release meeting our measurement criteria. The single-H200 suite therefore measures AR components within render-owned DyStream and Talker-T2AV (Table 7) and the UniLS motion baseline (Table 6). UniLS uses 0.15 s in Gen for motion prediction and 0.60 s in Post for FLAME decoding, rendering, smoothing, and export. Full-duplex X-Streamer and Wan-Streamer v0.2/v0.3 likewise separate a token-causal Thinker from an iterative visual Actor/Performer [83, 84, 229]. They inform causal scheduling while retaining diffusion visual ownership.

Autoregressive dialogue planning. Ex-Omni-2D connects autoregressive response planning to a flow-matching video generator [261]. Its dialogue model produces a structured visual plan, while generated speech units provide the shared acoustic timing. A distilled streaming variant carries the previous chunk's final clean latent into the next chunk. Under our visual-owner taxonomy, this is an AR-controlled diffusion pipeline. Its relevance here lies in causal planning and the transfer of motion context across generated chunks.

5.5 Fast Design Guidance

Accelerating visual autoregression targets the additive terms of Equation (5) through five design strategies:

- **Compress the serial visual state** ($C_{\mathrm{AR}}\downarrow$). Limiting per-frame representation size, such as MIDAS's 60 DC-AE tokens or UniMo's interleaved motion–video tokens, reduces quadratic attention cost while requiring the tokenizer and codec to preserve fine oral and facial details.
- **Parallelize local token emission** (serial depth $\downarrow$). Sampling all continuous tokens of a frame in parallel via a compact diffusion head (Body of Her) or a four-step local head (MIDAS), rather than executing dozens of sequential spatial steps, shortens serial latency while keeping local-head NFE and codec decoding on the critical path.
- **Bound and reuse persistent KV state** ($C_{\mathrm{AR}}(h_t)$ growth control). Truncating AR context to a fixed block window (MIDAS) or combining a sink-frame anchor with a sliding KV window and cyclic positions (EARTalking) caps per-step attention cost and memory footprint.
- **Decouple causal control from dense emission across devices** ($C_{\mathrm{system}}\downarrow$). As demonstrated in the diffusion-owned Wan-Streamer v0.2 pipeline, placing the lightweight token-causal Thinker on one GPU and running the heavy visual generator under Ulysses context parallelism across separate GPUs prevents control updates from stalling frame synthesis [83].
- **Explore speculative visual decoding.** Extending draft-and-verify speculative decoding from language models and general video AR [82, 111] to virtual humans offers a promising path to reduce serial calls, provided acceptance criteria preserve identity similarity and lip synchronization.

Reproducible AR benchmarks should report AR token counts, local-head NFE, KV cache update cost, codec time, and peak state memory over a unified RGB boundary.

5.6 Stable Design Guidance

Conditioning on committed visual tokens supports short-term continuity but can propagate errors once corrupted tokens enter the history. EARTalking combines an L1 reference sink, L3 cyclic re-indexing, and L4-C corrupted-history training. MIDAS uses L4-C training with separate eight-block AR and five-frame codec windows [28, 225]. None of the reviewed native visual-AR studies documents L2 multiscale memory, L5 motion-state handoff, or L4-G generated-history training with minute-scale drift curves. Matched long-session evaluation remains an open priority.

Table 6 Single-H200 runtime (s) and peak memory (GiB) for UniLS motion visualization. Stage, repetition, and attention-backend conventions follow Section 8.3.

Configuration	Year	N	$W \times H$	Attn.	Pre	Gen	Post	Load_1	Call $\pm s$	Warm peak M_w (GiB)
UniLS [38]	2026	32	512×512	SDPA_e	0.01	0.15	0.60	3.28	2.88 ± 0.98	3.73

Takeaways: Visual autoregression

Lower serial latency. Compress visual state and emit within-frame content in parallel, while checking codec errors on facial detail.

Higher throughput. Reuse KV state. Speculative visual decoding is a possible extension, but a draft model's identity fidelity remains unmeasured.

Minute-scale stability. Anchors and bounded history balance continuity against accumulated error and over-constrained motion. Generated-history training and adaptive refresh need matched long-session tests, including refresh discontinuities.

Hour-scale operation. Hierarchical state with explicit eviction, reset, and recovery policies is a research direction. Measure long-range forgetting, reset discontinuities, and memory and error trajectories.

5.7 Synthesis and Open Questions

Native visual autoregression offers a direct causal formulation for evolving human appearance, yet current systems demonstrate architectural feasibility more completely than standardized deployment speed or multi-minute stability. Co-designing high-compression visual tokenizers with low-NFE parallel emission heads, integrating sub-linear state-space or linear-attention backbones [65, 245] with explicit error recovery, and benchmarking complete RGB pipelines under time-binned drift protocols represent the key steps needed to make visual AR competitive with distilled diffusion and subject-specific rendering.

6 Render-Based Methods

6.1 Scope, Ownership, and Cost Boundary

Render-based pipelines synthesize video by animating a persistent, subject-specific appearance representation through compact time-varying controls. Under Section 2.3's rules, this route requires an explicit or implicit spatial scene asset to own appearance. Examples include a textured mesh, neural radiance field (NeRF), 3D Gaussian splatting (3DGS) cloud, or identity-conditioned image feature volume. The asset is geometrically queried, deformed, or rasterized to form RGB frames [68, 69, 99]. Whether the upstream controller predicts motion via a feed-forward network, an autoregressive model, or a diffusion/flow process does not alter this visual ownership.

Reusing an enrolled appearance asset couples computational efficiency directly to identity stability: amortizing appearance construction removes redundant per-frame texture synthesis while providing an architectural L1 identity anchor (Section 2.8). Nevertheless, producing each observable frame still incurs runtime costs across tracking, motion prediction, geometric deformation, rendering, and post-render composition. Over a session of T frames, total pipeline cost decomposes as

$$\begin{aligned} C_{\text{session}} = C_{\text{enroll}} + \sum_{t=1}^{T} \big[& C_{\text{track/cond}}(t) + C_{\text{motion}}(t) \\ & + C_{\text{repr}}(t) + C_{\text{render}}(t) \\ & + C_{\text{compose}}(t) + C_{\text{system}}(t) \big]. \end{aligned} \tag{6}$$

Here, C_{enroll} captures offline or one-shot subject preparation (multi-view calibration, monocular tracking, 3D reconstruction, rigging, per-identity optimization, and asset serialization). $C_{\text{track/cond}}$ covers runtime extraction of pose, camera, gaze, or audio features. C_{motion} covers predicting time-varying animation controls. C_{repr}, C_{render}, C_{compose}, and C_{system} cover spatial deformation or attribute updates, field integration or rasterization, neural shading/inpainting/paste-back, and media I/O. As shown in Figure 9, the recurring per-frame work depends on which of four rendering interfaces owns appearance: mesh/UV rasterization with neural shading, volumetric ray integration, 3D Gaussian splatting, or implicit image-space feature warping. Pioneering studio telepresence systems such as Codec Avatars demonstrated real-time person-specific rendering under dense multi-camera capture across deep appearance models, expressive facial rigs, and audio- or gaze-driven control [35, 132, 171]. Subsequent efforts reduce enrollment from multi-view studios to monocular video or a single portrait.

6.2 Explicit Controllable Appearance Representations

Explicit rendering backends bind appearance to interpretable geometric or volumetric structures driven by 3DMM, FLAME, SMPL, or SMPL-X parameters [9, 119, 134, 159], or accumulate dynamic texture and

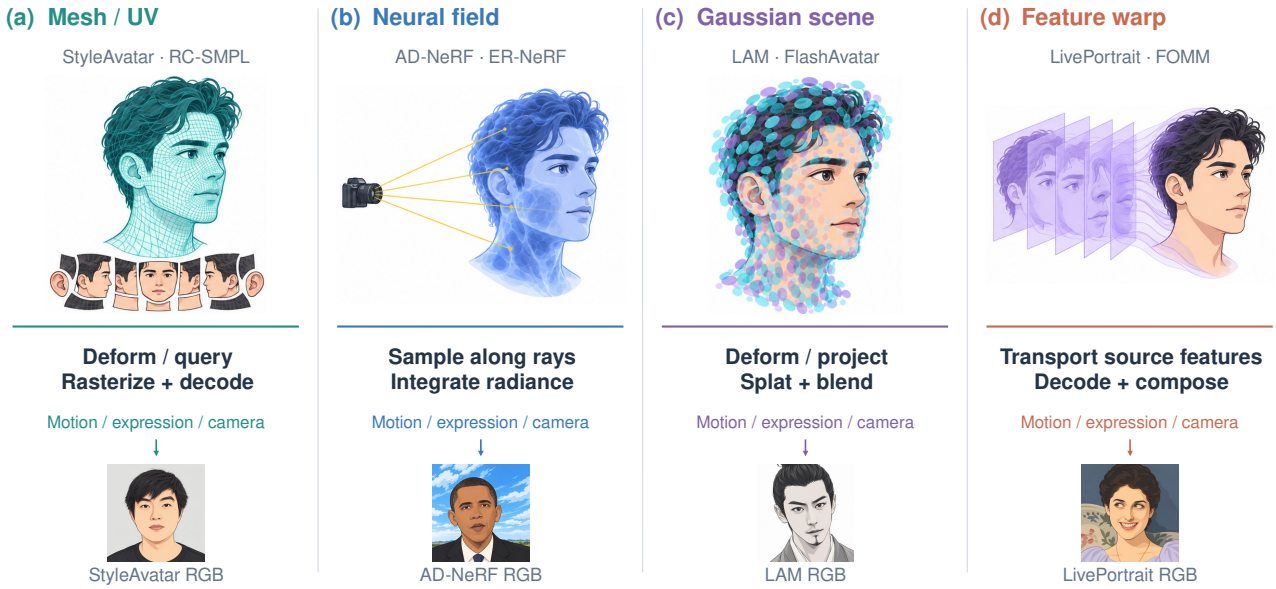


Fig. 9 Rendering interfaces: (a) mesh/UV appearance, (b) a neural field, (c) a Gaussian scene, and (d) feature transport. Representation sketches and bottom RGB portraits are schematic. StyleAvatar, AD-NeRF, LAM and LivePortrait exemplify the four interfaces, respectively [68, 69, 73, 212].

normal maps progressively from RGB-depth (RGBD) streams as in RC-SMPL [185]. RAM-Avatar rasterizes a personalized neural texture on a posed body template before its neural image decoder [51]; it belongs to this geometric-appearance branch despite adversarial decoder training. Within this paradigm, NeRFs and 3DGS represent the two dominant neural scene primitives [99, 149].

6.2.1 Neural-Field Backends

Early neural-field avatars established audio- and expression-driven volumetric rendering from monocular video at the cost of heavy per-ray multilayer perceptron (MLP) evaluations. AD-NeRF optimizes separate head and torso radiance fields [69]. HeadNeRF renders low-resolution volumetric feature maps followed by neural 2D upsampling [77]. FLAME-in-NeRF conditions volumetric queries on parametric expression coefficients [6]. To shrink ray-query cost (C_{render}) without sacrificing person-specific sharpness, subsequent methods adopt multi-resolution hash grids and occupancy pruning inspired by Instant-NGP [152]. RAD-NeRF decomposes spatial and audio queries into separate hash grids with a dedicated torso module [198]. ER-NeRF replaces 3D grids with compact tri-plane hash encoders and region-aware audio attention [117]. Building on tri-plane fields, SyncTalk refines the surrounding conditioning and composition stages via dedicated Face-Sync articulation alignment, Head-Sync pose stabilization, and Portrait-Sync hair/torso blending [162]. AvatarReX scales personalized neural fields to full-body, hand, and facial control using studio multi-view capture and surface-guided deferred rendering [283].

6.2.2 Gaussian Scene Representations

Replacing volumetric ray marching with differentiable tile rasterization of 3D Gaussians dramatically accelerates C_{render} while shifting design complexity toward primitive binding, non-rigid deformation (C_{repr}), and enrollment amortization (C_{enroll}). Mesh-bound Gaussian avatars attach primitives directly to parametric surface triangles. Yu et al.'s GaussianTalker binds Gaussians to FLAME, predicts speaker-specific blendshapes, and composites the rendered face back onto the source frame [253]. GaussianAvatars rigs multi-view Gaussians to FLAME triangles [166]. SplattingAvatar pairs mesh deformation with walkable barycentric Gaussians [177]. FlashAvatar embeds compact Gaussians on a UV surface with learned residual offsets [227]. Extensions of this rigged-Gaussian paradigm span blendshape-latent conditioning [54], personalized head geometry [279], SMPL-driven full-body animation [167]. Other extensions cover emotional speech control [241], progressive avatar construction [186], stylized toonification [187], and articulated hand modeling [276].

Direct audio-to-Gaussian deformation predicts primitive offsets from speech without explicit 3DMM blendshapes. Cho et al.'s GaussianTalker (distinct from Yu et

al.'s FLAME-bound method) uses spatial–audio cross-attention [32]. SyncTalk++ migrates tri-plane synchronization to dynamic Gaussians with out-of-distribution expression synthesis and torso restoration [161]. Richer deformation controllers predict expression-dependent Gaussian attributes in GaussianSpeech, decouple lip articulation from upper-face expression in DGTalker, adapt generalizable priors to target identities in GGTalker, and distill diffusion motion priors into full-body audio-driven Gaussians in AudioAvatar [4, 80, 110, 125]. To reduce per-frame deformation lookup cost (C_{repr}), EmbedTalk replaces tri-plane feature queries with per-Gaussian learned embeddings deformed by audio and facial controls [173]. GaussianHeadTalk demonstrates that inconsistent Gaussian-to-surface mapping and noisy upstream tracking can still induce visible temporal wobble despite a static identity asset [1].

Recent Gaussian pipelines push toward two deployment extremes: ultra-lightweight wearable execution and zero-shot feed-forward construction. For mobile and extended-reality (XR) headsets, TaoAvatar bakes non-rigid deformation into a compact network for Apple Vision Pro [26]. SqueezeMe eliminates runtime corrective MLPs to render multiple concurrent avatars on mobile processors [90]. Conversely, to eliminate hours of per-subject optimization ($C_{\text{enroll}}\downarrow$), generalizable builders reconstruct animatable assets in a single feed-forward pass. GPAvatar lifts one or more portraits into a controllable tri-plane field [37]. GAGAvatar reconstructs FLAME-rigged Gaussians via dual-lifting [36]. LAM, GUAVA, and VASA-3D predict animatable head or upper-body Gaussians directly from a single image [73, 234, 260]. SDTalk separates visible-surface Gaussian reconstruction from prior-guided completion of occluded oral interiors before applying coarse-to-fine motion fields [93]. While feed-forward builders reduce enrollment to seconds, unseen regions such as teeth and side profiles remain bounded by the capacity of the learned reconstruction prior.

6.3 Implicit Image-Space Appearance Reuse

Implicit image-space renderers warp a source portrait's 2D or 3D feature volume through learned motion correspondences, then decode it into RGB. Early one-shot methods include FOMM's local affine Jacobians and occlusion masks [183], articulated part warps [182, 184], and thin-plate-spline (TPS) transformations [275]. Face-vid2vid uses 3D keypoints, with later extensions for latent navigation and audio driving [216, 219, 220]. These methods rely on a convolutional decoder to infer dis-occluded backgrounds and mouth interiors. LivePortrait predicts canonical keypoints, head rotation, and expression residuals to warp the source volume before spatially-adaptive normalization (SPADE) decoding and paste-back [68]. EfficientSync selects pose-matched reference frames, warps and blends mouth features, and decodes the oral region with a SPADE generator [76]. Its spatial appearance reuse takes ownership precedence over adversarial training.

Spatial appearance transport also appears in adversarially trained systems. DINet deforms reference appearance features before mouth inpainting, while IP-LAP aligns reference images and multiscale features using predicted motion fields [271, 285]. EAT warps a source 3D appearance volume before decoding [63]. EDTalk disentangles mouth, pose, and emotion controls but uses a generator that spatially warps source features [194]. FixTalk retains that baseline generator while changing its motion and detail indicators [193]; this placement follows the paper's stated architectural inheritance rather than a separate implementation audit. These methods therefore belong to implicit appearance transport, not direct GAN synthesis.

6.4 Motion Controllers over Reused Appearance

Because implicit feature warpers and feed-forward Gaussian avatars expose low-dimensional driving interfaces, a broad class of recent methods trains expressive diffusion, flow-matching, or autoregressive controllers in compact motion space while delegating pixel synthesis to a frozen renderer. VASA-1 generates holistic facial dynamics, head pose, and gaze in a disentangled motion latent space before decoding a one-shot appearance volume [233]. FLOAT and FlowTalk apply rectified-flow matching to motion latents [50, 103]. Ditto structures controllable motion primitives [120]. ChatAnyone and MirrorMe extend compact motion generation to expressive upper- and half-body portraits [144, 165]. Similarly, MoDA, VibeAvatar, and Talk Like You pair multi-step or single-step flow/diffusion motion predictors (augmented with phonetic adaptation, aesthetic preference optimization, or personal speaking-habit conditioning) with frozen LIA or LivePortrait rendering modules [122, 205, 215]. Related motion–appearance factorizations appear in GAIA, JoyVASA, and style-conditioned talking heads [19, 72, 141, 217]. MambaTalk introduces selective state-space scanning for holistic gesture synthesis [236]. Pure speech-to-mesh predictors such as FaceFormer and Media2Face serve as upstream geometric controllers [58, 277].

In interactive and dyadic settings, causal motion controllers maintain conversational state over a persistent appearance asset. AVTR-1 predicts compact motion, refines driving keypoints, and warps cached source appear-

ance through a LivePortrait-based renderer [108]; its appearance backend is not a Gaussian scene. Teller drives keypoint warping from autoregressive quantized motion tokens [280]. ARIG combines framewise AR control, a local diffusion head, and a 512-slot interaction summary cache [70]. DyStream pairs AR/flow motion generation with a reference appearance volume [23]. REA-Listener drives GAGAvatar Gaussians with lightweight reactive control and background inpainting [36, 278]. EmbodiedHead predicts state-conditioned FLAME parameters for GAGAvatar rendering [269]. PortraitDirector and VASA-Rig target disentangled reenactment and virtual-reality (VR) rig control [91, 154].

6.5 Open-Source Measurements on One H200

Table 7 reports single-H200 timings across 26 render-route configurations under the protocol of Section 8.3. Gen spans both motion control and rendering when present. Pre-trained subject checkpoints or precomputed tracking are accepted where required by the repository. EDTalk, DINet, and EAT retain their models across calls; the other 23 scripts reload them. DINet spends 40.64 s in Pre versus 2.36 s in Gen, illustrating conditioning overhead even when landmarks are supplied offline. Stage decomposition exposes non-rendering costs from Equation (6). LAM spends 14.07 s in fresh FLAME tracking and preprocessing (Pre) compared to 0.97 s in core generation and rendering (Gen). PLY and media export in Post dominate GUAVA (50.92 s Post versus 9.45 s Gen) and SDTalk (33.92 s Post versus 12.42 s Gen). Note that SplattingAvatar and RAD-NeRF use their repositories' Python/PyTorch evaluation paths rather than custom Unity or CUDA runtimes. Ditto and Talk Like You execute overlapping threaded stages. SyncTalk runs native validation inside Call outside Pre/Gen/Post.

6.6 Fast Design Guidance

Across render-based pipelines, four recurring acceleration levers map to the cost terms in Equation (6):

1. **Amortize appearance construction at enrollment** ($C_{\text{repr}}, C_{\text{render}} \downarrow$ via C_{enroll}). Per-subject NeRFs and 3DGS assets invest minutes or hours of optimization for fast per-frame rendering, whereas one-shot feature warpers (LivePortrait) and feed-forward Gaussian builders (LAM, GAGAvatar) compress enrollment into a single forward pass at the cost of relying on learned priors for occluded regions.
2. **Replace volumetric ray queries with structured hashes or splatting** ($C_{\text{render}} \downarrow$). Multi-resolution hash grids (RAD-NeRF, ER-NeRF) and tile-rasterized 3D Gaussians with baked deformation networks (FlashAvatar, TaoAvatar, SqueezeMe) reduce rendering latency from seconds to milliseconds across desktop and mobile/XR processors.
3. **Run iterative sampling in compact motion space** ($C_{\text{motion}} \downarrow$). Diffusing low-dimensional motion latents or FLAME parameters (VASA-1, FLOAT, MoDA) keeps iterative network calls lightweight while leaving neural decoding and rendering on the critical path.
4. **Preserve static canvas pixels via paste-back composition** (C_{compose} trade-off). Compositing only the animated head or mouth onto the source frame avoids re-synthesizing backgrounds, requiring careful mask blending to prevent visible boundary seams.

6.7 Stable Design Guidance

While a persistent 3D asset or source feature volume provides an L1 identity anchor that limits recursive appearance synthesis, temporal failures still arise along the control, tracking, deformation, and composition stages. Recurrent or autoregressive motion controllers (such as ARIG and DyStream) can accumulate state drift over extended sessions, requiring explicit anchor conditioning in the motion controller to prevent gradual head-pose drift [23]. Independently of the controller, framewise tracking noise, out-of-distribution jaw articulations, and inconsistent Gaussian-to-triangle bindings introduce high-frequency geometric wobble and mouth-interior tearing, as analyzed in GaussianHeadTalk [1]. Evaluating render-based stability over multi-minute horizons therefore requires measuring pose drift, tracking wobble, disocclusion artifacts, and seam consistency alongside identity similarity.

6.8 Synthesis and Open Questions

Render-based pipelines excel when a subject's appearance can be enrolled once and animated repeatedly, scaling from high-throughput desktop splatting to mobile and XR headsets. However, separating appearance storage from motion control means neither enrollment cost nor tracking, deformation, and paste-back overhead can be omitted from runtime accounting, and static identity anchoring does not immunize a pipeline against controller drift or geometric wobble. Co-designing feed-forward 3DGS priors with history-robust

Table 7 Single-H200 runtime (s) and peak memory (GiB) for 26 render configurations. R: resident models. Stage, repetition, and attention-backend conventions follow Section 8.3.

Configuration	Year	N	$W \times H$	Attn.	Pre	Gen	Post	Load$_1$	Call $\pm s$	Warm peak M_w (GiB)
AD-NeRF [69]	2021	16	450×450	Eager	0.12	93.43	0.21	0.11	93.88 ± 1.60	2.12
RAD-NeRF [198]	2025	588	450×450	Eager	1.47	4.90	2.81	0.50	9.24 ± 0.04	1.24
ER-NeRF [117]	2023	16	450×450	Eager	0.11	0.19	0.42	2.61	1.69 ± 0.01	1.44
SyncTalk [162]	2024	16	512×512	Eager	7.36	0.22	0.35	0.59	14.05 ± 0.12	1.74
GPAvatar [37]	2024	16	512×512	Eager	1.12	0.77	0.92	3.48	4.14 ± 0.04	11.42
LAM [73]	2025	51	512×512	SDPA$_e$	14.07	0.97	0.96	23.09	31.95 ± 2.53	5.82
FOMM [183]	2019	169	256×256	Eager	3.19	1.19	0.46	2.07	5.52 ± 0.32	2.15
TPS Motion Model [275]	2022	169	256×256	Eager	3.45	1.45	0.45	1.94	6.02 ± 0.64	2.10
LivePortrait [68]	2024	78	600×704	Eager	2.15	1.26	2.34	36.37	8.20 ± 0.39	4.16
LIA [220]	2022	652	256×256	Eager	1.08	12.66	5.76	2.79	20.74 ± 1.59	4.71
EDTalk^R [194]	2024	251	256×256	Eager	0.92	4.64	1.70	2.74	7.27 ± 0.02	3.82
DINet^R [271]	2023	241	1920×1080	Eager	40.64	2.36	3.79	3.24	46.79 ± 0.11	1.85
EAT^R [63]	2023	88	256×256	Eager	0.04	2.01	0.51	3.67	2.56 ± 0.01	2.95
FLOAT [103]	2025	376	512×512	SDPA$_m$	0.03	4.14	6.17	7.61	16.04 ± 0.27	8.40
Ditto [120]	2025	394	1432×1432	SDPA$_e$	0.70	18.17	13.96	15.18	33.43 ± 2.44	9.42
MoDA [122]	2025	375	624×624	SDPA$_e$	0.37	7.29	6.89	47.21	25.43 ± 3.27	48.55
JoyVASA [19]	2024	321	512×512	SDPA$_e$	0.63	7.54	6.12	48.81	18.42 ± 2.01	7.12
DyStream [23]	2025	235	512×512	SDPA$_e$	0.66	16.38	3.70	19.88	30.99 ± 1.93	19.52
Talker-T2AV [249]	2026	136	512×512	FA2+SDPA$_f$	0.09	29.00	0.72	32.38	54.16 ± 0.96	20.48
Talk Like You [205]	2026	495	1034×1034	SDPA$_e$	4.46	12.52	10.92	21.71	35.24 ± 2.47	12.65
SplattingAvatar [177]	2024	16	1080×1080	Eager	0.27	1.10	0.58	3.77	4.68 ± 0.24	1.65
3DGS-Avatar [167]	2024	16	512×512	Eager	0.99	0.40	0.99	2.23	5.70 ± 0.88	0.71
GAGAvatar [36]	2024	241	512×512	SDPA$_e$	2.87	3.22	12.71	7.42	27.18 ± 1.09	2.29
GUAVA [260]	2025	425	512×512	xF(CUTLASS)	12.47	9.45	50.92	30.20	96.49 ± 4.51	6.03
SDTalk [93]	2026	744	512×512	SDPA$_e$	5.51	12.42	33.92	13.26	60.22 ± 2.15	2.67
face-vid2vid-unofficial [219]	2021	169	256×256	Eager	4.73	6.44	0.61	6.28	14.14 ± 1.14	4.24

Takeaways: Rendering-based generation

Repeated subject use. Gaussian assets and baked deformation amortize appearance construction. Account for capture requirements and hours of enrollment.

One-shot generality. LivePortrait warps, LAM Gaussians, and VASA-1 motion control support reuse from a source view. Test disocclusion, large poses, and mouth interiors under limited view coverage.

Mobile and wearable use. Splatting and baked correctives, as in TaoAvatar and SqueezeMe, support compact deployment. Per-subject assets remain necessary. Thermal and battery limits are untimed.

Minute-scale stability. An enrolled asset anchors appearance, but deformation and composition can still vary over time. Joint long-session reliability remains to be established.

Expressive control. Motion-space diffusion or flow can enrich control, within the asset's deformation space.

motion controllers and benchmarking complete tracking-to-composition pipelines over sustained sessions remain central open challenges.

7 Cross-Route Analysis and Design Trade-offs

Although GAN, diffusion, autoregressive, and render-based pipelines organize appearance synthesis and temporal state differently, they confront common system-level trade-offs when deployed under strict latency and stability budgets. This section synthesizes four cross-route architectural principles (concerning decoder selection, multi-GPU pipeline scheduling, state-management horizons, and non-identity failure paths) and connects them to the empirical findings of our single-H200 runtime benchmark and our matched-input HDTF quality benchmark.

7.1 Decoder Selection Reshapes the Speed and Fidelity Frontier

As algorithmic distillation reduces iterative denoising or motion prediction to a few steps, post-synthesis decoding

(D in Equation (2)) accounts for a growing fraction of per-chunk runtime. Substituting a pruned or lightweight decoder such as TinyVAE in place of a full-capacity variational autoencoder (as explored in PersonaLive and MotionStream [123, 181]) reduces decoding latency and raises throughput. It can also degrade identity similarity and worsen temporal perceptual consistency (measured via learned perceptual image patch similarity, LPIPS). Accelerating the visual decoder therefore alters the very identity and temporal stability properties that the upstream generator is trained to preserve. Decoder selection must therefore be evaluated jointly along both Fast and Stable axes.

7.2 Throughput and Startup Latency Follow Distinct System Paths

In streaming deployment, algorithmic distillation, multi-device parallelism, and kernel optimization affect steady-state throughput (FPS) and time to first frame (TTFF) along different system paths, as exemplified by the serving architecture of Live Avatar [87]. Reducing strict NFE through step distillation lowers both per-chunk work and initial response delay on a single device. By contrast, partitioning denoising steps across multiple GPUs via timestep pipeline parallelism increases steady-state chunk cadence, which is bounded by the slowest individual stage ($\max_i t_i$). It also adds inter-device pipeline-filling overhead to the initial traversal ($\sum_i t_i$). Achieving both high steady-state throughput and low startup latency therefore requires pairing timestep pipelining with dedicated VAE decoder parallelism and fused low-precision kernels.

Enrollment amortization shifts the break-even horizon. When comparing subject-specific renderers or personalized generators against zero-shot pipelines, one-time enrollment work C_{enroll} changes the effective per-frame cost over a session of N frames according to $\bar{C}(N) = C_{\text{enroll}}/N + C_{\text{runtime}}$. As an illustrative accounting example on a single reference GPU, an avatar requiring 3,600 GPU-seconds (1 hour) of enrollment and 0.02 GPU-seconds per rendered frame breaks even with a zero-shot generator costing 0.10 GPU-seconds per frame only after $N = 3{,}600/(0.10 - 0.02) = 45{,}000$ frames, or 30 minutes of generated video at 25 FPS. Deployment selection therefore depends jointly on expected session reuse, per-frame latency budgets, and whether one-shot appearance priors suffice for the target views.

7.3 Bounded Memory and Demonstrated Stability Are Distinct Properties

Architectural state bounds and empirical rollout stability address different deployment questions. In native visual AR and streaming diffusion, mechanisms such as sink-frame anchoring (L1), cyclic or Rolling RoPE position re-indexing (L3), and history-robust training (L4-C and L4-G, as in EARTalking and Live Avatar [87, 225]) curb error accumulation over their tested rollout windows. However, enforcing a fixed-capacity KV cache or memory bank establishes a hardware resource ceiling rather than guaranteeing freedom from visual drift, and survival under stationary or looped inputs does not automatically generalize to non-repeating interactive sessions with abrupt pose or scene transitions.

7.4 Appearance Reuse Leaves Non-Identity Failure Paths Active

By storing canonical texture and geometry in a persistent asset or source feature volume, render-based pipelines eliminate recursive synthesis of static identity details, yet remain vulnerable to temporal errors in motion prediction, tracking, deformation, and compositing. Without explicit anchor conditioning in a recurrent motion controller, head pose can drift progressively even when the underlying appearance volume is fixed, as studied in DyStream [23]. Similarly, framewise tracking noise and inconsistent Gaussian-to-mesh bindings can induce high-frequency geometric wobble and oral artifacts, as analyzed in GaussianHeadTalk [1]. Architectural appearance reuse thus narrows the surface of visual drift to the control and deformation path, motivating the transfer of L4 history-robust training and TaoMate-style reliability gating [62] to recurrent motion controllers.

7.5 Empirical Insights from Our Runtime and Quality Benchmarks

Our two empirical studies reinforce these cross-route principles with direct measurements. First, in our single-H200 runtime benchmark across 51 open-source configurations (Tables 4, 5, 7, and 6, Section 8.3), stage decomposition exposes where wall-clock time is spent outside core synthesis. In warmed execution, MuseTalk spends 0.85 s in core latent generation (Gen) inside a 41.53 s call dominated by preprocessing (20.68 s) and blending/encoding (19.99 s). Fresh tracking and preprocessing dominate LAM (14.07 s Pre vs. 0.97 s Gen) and DINet (40.64 s Pre vs. 2.36 s Gen). Mesh/media export dominates GUAVA (50.92 s Post vs. 9.45 s Gen), and

41 of 51 configurations reload weights on every invocation (Figure 10). Second, in our matched-input benchmark of 23 audio-driven configurations across 40 fifteen-second HDTF cases (920 generated videos, Table 9 and Section 8.4), the routes have different strengths. Regional GAN editors (Wav2Lip+GAN, VideoReTalking) and sync-supervised diffusion models (SoulX-FlashTalk, SoulX-LiveAct, InfiniteTalk) lead in SyncNet confidence. Appearance-warping and regional editors (Ditto, MuseTalk, Wav2Lip+GAN, EDTalk, MoDA) achieve the highest reference ArcFace cosine similarity. Full-frame diffusion backbones (Live Avatar, StableAvatar, InfiniteTalk, SoulX-FlashHead Pro) achieve the lowest full-frame FVD and FID. Within these benchmarks, no route leads across all measured dimensions.

8 Datasets and Evaluation

8.1 Datasets for Generation and Temporal Evaluation

Table 8 summarizes representative corpora used to train and evaluate virtual-human video generators. Single-speaker facial datasets serve distinct tasks: HDTF provides high-definition talking portraits for audio-driven lip synchronization [272], VFHQ targets video-face super-resolution [228], CelebV-HQ supplies attribute annotations for portrait generation and editing [289], and MEAD captures multi-view emotional speech across 60 actors [210]. At web pretraining scales, TalkVerse [222], OpenHumanVid [116], and EchoMimicV2 [147] curate millions of synchronized audiovisual and pose-annotated clips across portrait, half-body, and full-body scopes.

For interactive and 3D reconstruction settings, ViCo [287] and ResponseNet [137] pair speaker streams with listener responses, while SpeakerVid-5M [267] and Seamless Interaction [2] scale single-speaker and dyadic dialogue curation to thousands of hours. NeRSemble provides calibrated 16-view high-resolution head recordings for dynamic 3D avatar reconstruction [105], and recent benchmarks further incorporate preceding context or scene-level human–object interactions, including Face-to-Face [34], InteracVid [259], and the OmniHuman dataset with OHBench [126,291]. Aggregate corpus hours mostly consist of short 3–50 s clips rather than uninterrupted multi-minute sessions. Live Avatar, for example, uses looped audio for extended tests [87]. Dataset duration must therefore match the target task.

Table 8 Summary of representative virtual-human datasets.

Dataset	Year	Scale and media
MEAD [210]	2020	281.4k clips, 60 actors, 7 views, 1080p, 30 FPS
HDTF [272]	2021	362 videos, 15.8 h, 300+ subjects, 720p/1080p
VFHQ [228]	2022	16,827 clips, 7,228 videos, 512^2–1000 px faces
CelebV-HQ [289]	2022	35,666 clips, ∼65–68 h, 15,653 subjects, $\geq 512^2$
ViCo [287]	2022	483 dyadic pairs, 95.4 min, 92 subjects, 384^2
NeRSemble [105]	2023	4,734 sequences, 7.5 h, 222 subjects, 16 views, 73 FPS
ResponseNet [137]	2025	696 dyadic pairs, 14.2 h, 161 subjects, 1024^2
TalkVerse [222]	2025	2.3M clips, 6.3k h, 720p/1080p, 25 FPS
OpenHumanVid [116]	2025	13.2M clips with caption, skeleton, and audio
Seamless Inter. [2]	2025	64,739 dyadic interactions, 4,065 h, 4,284 subjects, 4K
SpeakerVid-5M [267]	2026	5.2M single (8.7k h) and 770k dialogue (1.8k h) clips

8.2 Evaluation Metrics and Protocols

Evaluating virtual-human generation combines visual quality, human-specific fidelity, system timing, and longitudinal stability. For paired reconstruction, Peak Signal-to-Noise Ratio (PSNR), the Structural Similarity Index (SSIM) [78,221], and Learned Perceptual Image Patch Similarity (LPIPS) [100,264] measure framewise fidelity. Unpaired distributional realism is assessed via Fréchet Inception Distance (FID) [74], spatiotemporal Fréchet Video Distance (FVD) on Inflated 3D ConvNet (I3D) features [203], and CLIP similarity [168]. Human-specific evaluators measure identity cosine similarity (CSIM) in ArcFace space [49], audiovisual synchronization via SyncNet Lip-Sync Error Distance (LSE-D) and Confidence (LSE-C) [39,164], and paired mouth-keypoint error via Landmark Distance (LMD) [27]. Complementary measures include soft dynamic time warping (Soft-DTW) for phase-tolerant motion alignment [273], cross-clip oral references in RGOR for teeth fidelity [197], whole-body/tracking wobble and human-preference evaluators such as HVEval+ [1,38,110,226].

For deployment and long-horizon stability, runtime reports must separate enrollment from inference and pair throughput (FPS, RTF) with time to first frame (TTFF), inter-chunk jitter, strict NFE, and peak GPU memory (Section 2.7). Longitudinal evaluations should track quality, identity, synchronization, and resource usage across predeclared time bins (for example, 0–10 s, 10–60 s, and multi-minute windows) rather than relying on single session-wide averages, logging detection failures, resets, and recovery alongside structured human studies.

8.3 Single-H200 Runtime and Memory Measurements

To evaluate stage-resolved runtime and memory costs under a unified hardware environment, we benchmark

51 open-source configurations (spanning 8 GAN, 16 diffusion, and 26 render-based pipelines plus one UniLS FLAME motion-visualization baseline) on a single NVIDIA H200 GPU (Tables 4, 5, 7, and 6). We include only configurations with complete first-call loading, warmed stage timings, and GPU memory records. No native visual-AR repository met these inclusion criteria.

Repetition protocol and model residency. Each configuration executes three warm-up and three measured calls in one Python interpreter. Every call replays the same input clip and generates a fresh output. Load_1 records loading during the first warm-up. Pre, Gen, Post, and Call report means across the three measured calls, with sample standard deviation s for Call. Ten configurations (marked R) retain model objects while repeating preprocessing and state initialization. The remaining 41 reload weights on every call. All six per-call records and stage statistics are available in the companion archive.

Hardware isolation and workload heterogeneity. Each suite uses one H200, four assigned CPU-affinity identifiers, and a four-thread limit for numerical libraries. Suites run across three host nodes. Runs affected by foreign GPU processes or cross-device leakage are rejected. Repositories differ in demo clips, sampling schedules, precision, offloading, and precomputed assets. The tables report actual frame counts N and canvas dimensions $W \times H$, excluding comparison panels. Required landmarks, audio features, and subject-trained checkpoints remain supplied inputs, so their offline preparation is excluded. LAM performs fresh tracking during Pre.

Inputs and implementations. Unless otherwise stated, we use the demo inputs recommended in each official open-source repository's README. Wav2Lip+GAN uses the EDTalk demo pair, and FOMM uses the TPS Motion Model demo pair. Moore-AnimateAnyone and face-vid2vid-unofficial use third-party implementations and their supplied demos. RAD-NeRF uses its PyTorch reimplementation. Per-configuration sampling settings and stage definitions are retained in the companion archive.

Attention backends. The Attn. column records backend types. FA2 and FA3 denote FlashAttention 2 and 3, and xF denotes xFormers. SDPA subscripts m, e, f, and c indicate math, memory-efficient, flash, and cuDNN paths. Eager indicates that no fused attention API was reported. Superscript S marks seven labels obtained from source inspection. The other 44 derive from prior execution observations.

Stage boundaries and memory accounting. Stage timers use monotonic clocks synchronized at Torch CUDA or Jittor boundaries. Pre captures input preparation and condition encoding. Gen captures the primary neural synthesis or rendering path. Post captures decoding (including dense-diffusion VAE decoding and UniLS FLAME visualization), paste-back composition, and video encoding. Appearance-reuse pipelines generally include both motion controller and renderer inside Gen and FLOAT retains its internal encoders in Gen. Call spans invocation entry to return, including reloading, SyncTalk validation, and threaded overlap in Ditto or Talk Like You. Load_1 therefore cannot be summed with warmed Pre/Gen/Post to obtain Call. Figure 10 illustrates first-call stage profiles for MuseTalk, LAM, and GUAVA. Peak warmed memory M_w records the maximum NVIDIA Management Library (NVML) device memory sampled at roughly 100 ms intervals across all suite processes during the three measured calls, complemented by per-call Torch allocator counters in the archive.

Provenance and interpretation boundary. We reconciled all reported Load_1 values, warmed stage means, sample standard deviations, frame counts, canvas dimensions, and M_w peaks against the 306 archived iteration records and decoded the primary output video of every configuration to verify frame count and resolution. Because these 51 runtime configurations execute each repository's native demo workload and residency mode over short clips, they expose within-pipeline stage bottlenecks rather than establishing a cross-method ranking or continuous-session stability. Section 8.4 complements them with a matched-input quality benchmark across 23 audio-driven configurations.

8.4 Matched-Input Audio-Driven Benchmark on an HDTF Subset

Because the single-H200 runtime suites in Section 8.3 execute each repository's native demo inputs, they isolate within-pipeline stage costs without comparing output quality on shared inputs. To evaluate visual fidelity, identity preservation, and lip synchronization under a unified input protocol, we conduct a matched-input benchmark across 23 audio-driven configurations on a 40-case subset of HDTF [272] following the evaluation split of Let Them Talk [107], yielding $23 \times 40 = 920$ generated videos (600 s per configuration, Table 9). The cohort includes four GAN generators: Wav2Lip+GAN, MuseTalk, VideoReTalking, and MakeItTalk. It also includes 12 diffusion generators

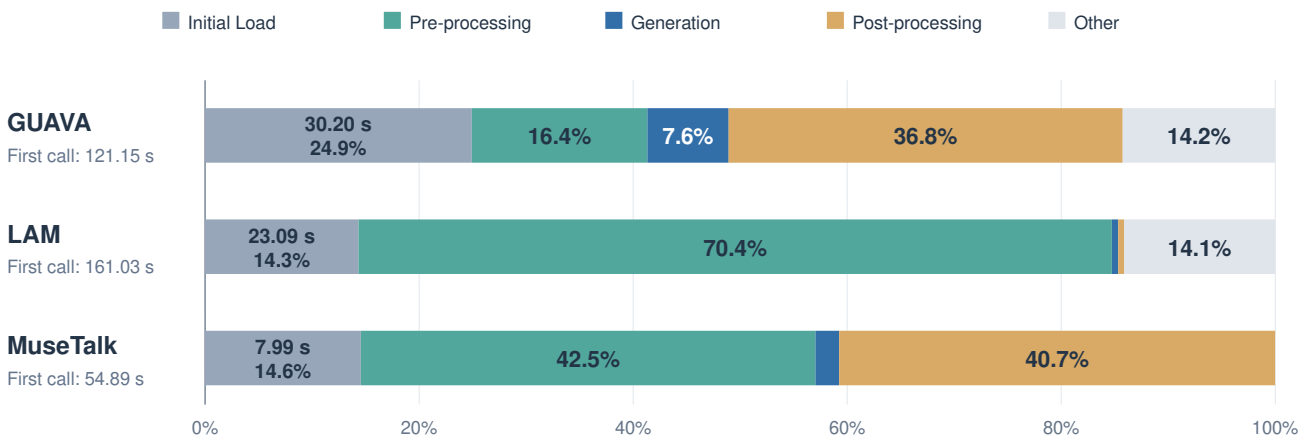


Fig. 10 First-call stage composition from our single-H200 measurements. All stages and the Call denominator use the same initial invocation. Load labels give seconds and percentage. Other is the elapsed-time residual after the four recorded phases. The main tables use warmed means for Pre/Gen/Post/Call. Workloads and residency differ.

(StableAvatar, InfiniteTalk, LiveTalk, AvatarForcing-SW, LeapTalk, Avatar-Forever, SoulX-LiveAct, SoulX-FlashTalk, SoulX-FlashHead Lite, SoulX-FlashHead Pro, LightningCP, and Live Avatar). Seven rendering pipelines reuse appearance: EDTalk, FLOAT, Ditto, MoDA, JoyVASA, DyStream, and Talk Like You.

Input cases and evaluation protocol. Each of the 40 test cases from the Let Them Talk HDTF split [107] pairs a single reference portrait frame with a 15-second, 16-kHz mono driving audio track. Configurations that accept text conditioning additionally receive the case's scene description. Native output frame rates, canvas dimensions, and command-line invocations are preserved in the per-case archive. For MakeItTalk, we extract only the central generated portrait from its native triptych. Every output video is scored under a fixed three-part protocol:

1. **Audiovisual synchronization (Sync-C ↑, Sync-D ↓).** Lip-sync confidence and distance are computed using the official `syncnet_python` pipeline with S3FD face tracking and cropping at 25 FPS, 16 kHz mono audio, and a temporal search window of `vshift` = 15, without pre-scoring offset alignment.
2. **Identity preservation (ID ↑).** Identity retention is measured using InsightFace 0.7.3 (`buffalo_l`: `det_10g.onnx` detector and `w600k_r50.onnx` ArcFace recognizer) as raw cosine similarity in $[-1, 1]$ (without $(\cos+1)/2$ rescaling) against each case's reference portrait. For every clip, 16 deterministic temporal-bin midpoint frames (excluding frame 0) are sampled, averaged within each clip, and then averaged across the 40 clips. To prevent tight portrait crops from failing detection without altering facial pixels, both reference and generated frames apply a uniform detection fallback (original frame, then 10%, 20%, and 40% black-border padding at a 0.5 confidence threshold).
3. **Unpaired visual and spatiotemporal distribution quality (FID ↓, FVD ↓).** Because the 40 test cases cross-pair reference portraits with driving audio tracks, pixel-aligned ground-truth videos do not exist. We therefore compute model-level FID and FVD between each configuration's 40 generated 15-second videos and an in-domain reference pool of 20 real 15-second HDTF clips (25 FPS), using full-frame aspect-preserving center-square RGB crops without face-detection filtering. FID samples 96 uniform frames per clip resized to 256×256 using `pytorch-fid` 2048-dimensional Inception features. FVD samples four stratified contiguous 16-frame windows per clip resized to 224×224 (scaled to $[-1, 1]$) using the StyleGAN-V TorchScript 400-dimensional I3D backbone.

Empirical findings and interpretation boundaries. Table 9 reveals distinct capability profiles across the three evaluated routes rather than a single monolithic winner:

– **Synchronization vs. full-frame distributional realism.** Regional GAN editors trained with explicit lip-sync critics achieve the highest SyncNet confidence and lowest distance (Wav2Lip+GAN at 7.8614 Sync-C and VideoReTalking at 7.1019 Sync-D). Other high-confidence results come from sync-supervised streaming and multi-step diffusion systems (SoulX-FlashTalk at 7.6845, SoulX-LiveAct at 7.6385, InfiniteTalk at 7.4545, and SoulX-FlashHead Lite at 7.1473) and rectified-flow motion controllers (MoDA at 7.0538). Conversely, full-frame diffusion backbones achieve the lowest spatiotemporal FVD and image FID against the real HDTF reference

Table 9 Matched-input evaluation of 23 audio-driven configurations across 40 fifteen-second HDTF test cases (920 generated videos) using the benchmark split of Let Them Talk [107]. Sync-C and Sync-D are SyncNet confidence and distance at 25 FPS without pre-scoring offset correction, ID is raw InsightFace ArcFace cosine similarity across 16 temporal-bin midpoints per clip, FID and FVD compare 40 generated center-square full-frame clips against 20 in-domain real 15-second HDTF reference clips. Protocol and interpretation boundaries: Section 8.4.

Configuration	Year	Route	Sync-C ↑	Sync-D ↓	ID ↑	FID ↓	FVD ↓
Wav2Lip+GAN [164]	2020	GAN	7.8614	7.2206	0.9345	125.5026	460.0848
MuseTalk [268]	2024	GAN	6.7746	7.6902	0.9402	123.2035	460.0232
VideoReTalking [30]	2022	GAN	7.5760	7.1019	0.8726	122.6412	456.9658
MakeItTalk [288]	2020	GAN	4.4430	9.5825	0.8947	118.5316	439.5050
StableAvatar [202]	2025	Diffusion	3.5585	11.4132	0.8549	113.8779	383.7462
InfiniteTalk [243]	2025	Diffusion	7.4545	7.7165	0.8629	113.9264	389.5934
LiveTalk [31]	2025	Diffusion	3.5085	10.9031	0.8016	118.1108	424.3754
AvatarForcing-SW [46]	2026	Diffusion	4.0755	11.1357	0.7680	116.9224	419.1612
LeapTalk [265]	2026	Diffusion	6.9870	8.2025	0.8668	120.2615	405.0797
EDTalk [194]	2024	Render	6.2825	8.8483	0.9263	123.6688	389.8324
FLOAT [103]	2025	Render	6.1561	8.6035	0.9091	143.3191	402.5805
Ditto [120]	2025	Render	5.3529	9.3416	0.9647	120.3729	412.5909
MoDA [122]	2025	Render	7.0538	7.9658	0.9261	118.0633	395.3063
JoyVASA [19]	2024	Render	6.4922	8.6513	0.8942	117.9605	418.7080
DyStream [23]	2025	Render	6.9169	7.9157	0.9237	133.1542	523.6795
Talk Like You [205]	2026	Render	6.5973	8.0582	0.8851	121.4178	402.9893
Avatar-Forever [118]	2026	Diffusion	3.6331	11.3872	0.8923	147.6847	767.2000
SoulX-LiveAct [281]	2026	Diffusion	7.6385	7.6401	0.8696	116.6186	395.7511
SoulX-FlashTalk [179]	2025	Diffusion	7.6845	8.2497	0.9059	169.2544	1199.6986
SoulX-FlashHead Lite [256]	2026	Diffusion	7.1473	7.9148	0.9019	115.4446	413.4087
SoulX-FlashHead Pro [256]	2026	Diffusion	6.4332	8.6539	0.9259	115.6840	390.0262
LightningCP [133]	2026	Diffusion	6.8113	7.9428	0.9163	120.0276	419.8881
Live Avatar [87]	2025	Diffusion	6.6594	8.6303	0.8445	117.3549	373.4172

pool, led by Live Avatar (373.42 FVD), StableAvatar (113.88 FID, 383.75 FVD), InfiniteTalk (113.93 FID, 389.59 FVD), and SoulX-FlashHead Pro (115.68 FID, 390.03 FVD), alongside EDTalk (389.83 FVD) and MoDA (395.31 FVD).

– **Structural scope shapes identity cosine similarity.** The highest ArcFace similarities occur in three types of pipeline. Appearance warpers retain a persistent source representation (Ditto at 0.9647, MoDA at 0.9261, and DyStream at 0.9237). Regional editors modify the lower face and retain surrounding source pixels (MuseTalk at 0.9402 and Wav2Lip+GAN at 0.9345). EDTalk constrains head pose and scores 0.9263. SoulX-FlashHead Pro and LightningCP also achieve high similarity, at 0.9259 and 0.9163, respectively. High ID scores in mouth-crop or fixed-pose editors reflect preserved source pixels as much as generative fidelity, whereas dense diffusion models that synthesize larger head rotations and upper-body motion naturally exhibit wider embedding variation.
– **Sensitivity of full-frame FID and FVD to native framing.** Because FID and FVD are evaluated on full-frame center-square crops without face re-cropping, they penalize differences in native portrait framing, aspect ratio, or background handling. For example, SoulX-FlashTalk achieves top-tier lip synchronization (7.6845 Sync-C) and strong identity preservation (0.9059 ID), yet records higher full-frame FID (169.25) and FVD (1199.70). These scores are sensitive to its native framing relative to the reference pool. Moreover, because the real reference pool contains 20 clips, small distributional differences (such as the 0.05 FID gap between StableAvatar and InfiniteTalk) should be interpreted cautiously given finite-sample variability. They do not establish a reliable composite ranking.

8.5 Streaming Evaluation

We evaluate six configurations on a 1/5 (20%) subset of StreamEval's avatar-performance conditions [199]: LiveAvatar, SoulX-LiveAct, Avatar-Forever, SoulX-FlashTalk, AvatarForcing, and LiveTalk. Each receives an initialization portrait, incrementally released speech, and live visual directions. Three repetitions per condition give 36 sessions per configuration and 216 sessions in total.

The subset covers all five duration strata: 15, 30, 60, 120, and 300 s. Selection uses benchmark metadata, independently of generated outputs and scores. The subset

Table 10 Streaming avatar-performance results with local Qwen judges. All scores use a 0–100 scale (↑).

Model	Quality	Stability	Following	Identity
Avatar-Forever	24.76	31.09	14.94	65.01
AvatarForcing	54.38	66.06	8.89	88.97
LiveAvatar	25.55	30.33	12.59	88.25
LiveTalk	40.70	39.00	12.22	82.26
SoulX-FlashTalk	44.53	49.20	0.00	97.00
SoulX-LiveAct	22.92	26.94	13.09	70.23

was fixed before judging, and matching sessions already collected were retained unchanged. The first started attempt is retained, including failures, and unsupported visual controls remain in the Following denominator. Aggregation averages repetitions and conditions within each duration stratum before taking the equal-weight mean across the five strata.

Table 10 reports four StreamEval content metrics: Quality, Stability, Following, and Identity. Synchronization evidence remains unresolved, so we omit A–V Sync and the dependent aggregate scores. All six configurations support custom portraits. StreamEval assigns five points for this capability, regardless of content quality. This constant bonus does not distinguish the tested configurations and is omitted from the table.

Locally deployed Qwen vision models replace StreamEval's reference judges, and independent calibration has not been established. The subset and judge changes limit direct comparison with published StreamEval scores. On the tested subset, AvatarForcing has the highest Quality and Stability estimates (54.38 and 66.06), while SoulX-FlashTalk has the highest Identity estimate (97.00). Following scores range from 0.00 to 14.94.

9 Discussion, Challenges, and Future Directions

The four visual-ownership routes distribute appearance storage, motion control, and temporal state across different stages of the generation pipeline. The analyses in Sections 3–7 identify four challenges: specifying modular interface contracts in hybrid pipelines, standardizing complete-system cost accounting, sustaining bounded-state generation over long interactive horizons, and integrating provenance and security safeguards within real-time latency budgets.

9.1 Modular Composition Requires Explicit Interface Contracts

Modern virtual-human systems increasingly operate as hybrid cascades that couple multimodal planners or motion generators with separate visual owners, decoders, and compositors (Section 2.3, Table 2) [23, 70]. While modular factorization allows researchers to upgrade a motion predictor or swap a VAE decoder independently, our analysis shows that local improvements frequently expose upstream or downstream bottlenecks, such as audio look-ahead buffering in Teller or feature extraction and paste-back blending in LivePortrait [68, 280].

Making hybrid pipelines predictable in both latency and long-horizon stability requires formalizing explicit interface contracts across modules. Each stage should declare its spatial coordinate frame, identity reference format, prerequisite subject assets, retained temporal state, and error-reset semantics. Low-dimensional parametric or implicit motion interfaces reduce the state complexity seen by a renderer at the cost of restricting topological changes and fine oral details, whereas dense latent interfaces preserve open-domain expressiveness while coupling identity retention to temporal state updates. Establishing typed representation, timing, and recovery contracts so that each module's latency and drift contribution can be verified both in isolation and in composition remains a foundational systems challenge.

9.2 Complete-System Cost Accounting

Across the reviewed literature, reported FPS, latency, RTF, and NFE figures frequently measure incompatible subgraphs, such as denoiser-only calls or model-only rendering without tracking and composition [68, 87]. Our single-H200 benchmark of 51 configurations demonstrates that when the complete call boundary is timed, preprocessing (as in DINet and LAM), postprocessing/export (as in MuseTalk, PC-AVS, and GUAVA), and per-call model reloading (present in 41 of 51 repositories) often dwarf core neural generation. Future benchmarks should therefore separate one-time enrollment (C_{enroll}) from runtime execution, maintain persistent model residency, and report stage-resolved profiles covering input buffering, conditioning, core synthesis, decoding/rendering, and I/O alongside user-facing TTFF, inter-chunk jitter, and peak memory as resolution, chunk length, and session duration scale.

Table 11 Requirements that guide method selection. Suitability depends on the intended task, available resources, and supported controls.

Requirement	Candidate mechanism	Additional condition to check
Regional lip editing	Wav2Lip+GAN, MuseTalk	Source pixels remain usable, inspect seams, synchronization, and jitter.
Reusable identity and controllable pose	Fields, Gaussians, feature warps	Enrollment is feasible *and* the representation supports the intended appearance, views, and motion.
One-shot portrait animation	LivePortrait, VASA-1, LAM	Test disocclusion, pose limits, and complete controller–renderer timing.
Appearance or scene changes	Dense visual generation	Validate the specific change. A static-scene system such as Live Avatar does not establish open-ended scene editing.
Explicit visual-history control	Native visual AR	Profile local sampling and decoding, test long-rollout errors and resets.

9.3 Bounded-State Long-Horizon Generation

While the L1–L5 mechanisms in Section 2.8 provide a unified toolkit for curbing temporal degradation, their empirical validation remains uneven across routes. Streaming diffusion pipelines have demonstrated combinations of all five mechanisms, including Live Avatar's multi-hour looped-audio rollout [87]. Render-based systems rely on an architectural L1 appearance asset but remain susceptible to controller pose drift (as studied in DyStream [23]) and tracking wobble. Native visual AR models lack standardized multi-minute drift curves. Stateless GAN editors remain prone to framewise jitter.

Bridging these gaps calls for cross-route transfer and controlled ablation of state-management policies. On-policy generated-history training (L4-G), anchor-gated memory updates (as in TaoMate [62]), and diversity-promoted long-term banks (as in InteractiveAvatar [188]) offer generalizable strategies that can be evaluated in native visual AR and recurrent motion controllers as well as diffusion. Crucially, as shown by motion damping in self-forced diffusion and its mitigation in DynaForcing and Routed Forcing [88,191], long-horizon benchmarks must evaluate identity preservation, lip synchronization, motion amplitude/diversity, and responsiveness to intentional scene or reference switches jointly across predeclared time bins.

Direct audio-to-deformation and multidimensional stability. Recent 3D Gaussian pipelines increasingly bypass intermediate 3DMM or SMPL parameters in favor of direct audio-to-Gaussian deformation fields, preserving high-frequency facial, hand, and body dynamics while shifting complexity into learned deformation networks and upstream trackers [1,4,80,110,125]. Because fixed Gaussian identity assets do not prevent tracking jitter, primitive tearing, or out-of-distribution articulation failures, evaluating these pipelines requires pairing enrollment and runtime timings with time-binned measurements of geometric wobble, synchronization, and recovery.

9.4 Route Selection Under Deployment Constraints

Synthesizing the route-specific guidance from Sections 3–6, Table 11 maps common application requirements to candidate architectural routes and their primary verification checks:

- **When per-subject enrollment is feasible**, render-based 3DGS and neural-field avatars offer high runtime throughput and compact mobile or XR execution [26,227], provided capture and fitting costs (C_{enroll}) are amortized over sufficient session duration and tracking/compositing stability is verified.
- **When zero-shot one-shot portrait animation is required**, pairing a compact motion generator or keypoint predictor with implicit feature warping or feed-forward 3DGS construction provides an effective balance of generality and speed [68,73,233], bounded by source-view disocclusion and extreme pose limits.
- **When synthesis must accommodate broad appearance, clothing, or scene changes**, distilled dense diffusion with rolling KV memory and multi-GPU pipelining supports open-domain visual generation [87,179], requiring careful tuning of reference anchors so they do not suppress intended scene transitions.
- **When only localized lip dubbing is needed on an existing video**, feed-forward regional GANs minimize generator cost [164,268], shifting optimization to fast face preparation, seam blending, and temporal smoothing.
- **When explicit causal modeling of dense visual state is central**, native visual AR provides a unified sequence formulation whose local emission overhead and long-rollout recovery should be profiled end to end before deployment.

9.5 Trustworthy Deployment

Advances in Fast and Stable generation heighten misuse risks: low-latency synthesis enables real-time voice- and

video-driven impersonation, while persistent identity anchoring sustains deceptive sessions over extended durations. Addressing these risks requires layered technical and governance controls rather than a single scalar score. The National Institute of Standards and Technology (NIST) AI 100-4 framework distinguishes provenance tracking (cryptographically binding content metadata and watermarks at creation) from post-hoc synthetic-content detection [21]. For live streaming media, the Coalition for Content Provenance and Authenticity (C2PA) specification defines segment-level cryptographic manifests that bind signing credentials to media bytes and expose tampering [42]. Manifests can be stripped by uncooperative platforms and do not by themselves verify subject consent or authorization to animate a likeness [40, 41]. Complementing cryptographic provenance, passive detectors benchmarked in DeepfakeBench still exhibit cross-manipulation generalization gaps [240], while imperceptible watermarking systems such as Video Seal achieve faster-than-playback embedding under offline boundaries but require further hardening against adaptive removal attacks [60]. Crucially, in real-time interactive deployment, consent verification, watermark embedding, and segment-level C2PA signing execute on the live output path and must therefore be budgeted directly inside the per-chunk latency decomposition alongside decoding and I/O.

9.6 Conclusion

Fast and Stable virtual-human generation depends on appearance representation, computation scheduling, and temporal state. The visual-owner taxonomy clarifies how feed-forward GANs, diffusion and flow models, native visual autoregression, and renderers allocate enrollment and runtime costs. It also exposes their different sources of error accumulation. We measured 51 single-H200 configurations and evaluated 23 audio-driven configurations on 40 fifteen-second HDTF cases (920 videos). These studies show that end-to-end performance is governed as much by conditioning, decoding, model residency, and composition as by core network calls. High identity similarity or bounded memory alone establishes neither full-frame realism nor long-horizon stability. Future benchmarks should jointly measure service latency, stage costs, and stability over time under diverse interactive inputs.

Data Availability Statement

The runtime release contains 56 single-H200 configurations (336 calls, with 51 configurations reported in Section 8.3). It includes summaries, stage records, provenance, NVML traces, and videos. The HDTF quality release (Section 8.4) contains 920 generated videos (23 configurations × 40 cases), 940 feature/metric records, inputs, and evaluation scripts. The public companion provides the measurements, source records for the measured image crops, and a searchable literature catalogue.

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